

ESAT PRACTICE TEST SERIES

PHYSICS + BIOLOGY

LEVEL 2 • HARDER THAN ESAT

Physics + Biology

Mathematics 1 + Physics + Biology

Challenge material with denser interpretation, quantitative reasoning and longer solution chains.

MODULE

Mathematics 1

MODULE

Physics

MODULE

Biology

FULL SOLUTION BOOK

ThrivingScholars

81 preserved questions • three verified module keys

Physics + Biology · Level 2 Harder than ESAT

This book compiles three existing 27-question modules into one 81-question pathway. No new questions have been generated and no source solution has been rewritten.

Composition: Mathematics 1 + Physics + Biology. Each module retains its original question order, answer and worked solution.

Mathematics 1

Engineering Mock Test 5 · preserved Mathematics 1 module

Q1 C	Q2 B	Q3 D	Q4 A	Q5 E
Q6 C	Q7 B	Q8 D	Q9 A	Q10 E
Q11 C	Q12 B	Q13 D	Q14 A	Q15 E
Q16 B	Q17 C	Q18 D	Q19 A	Q20 E
Q21 C	Q22 B	Q23 D	Q24 A	Q25 E
Q26 C	Q27 B			

Physics

Engineering Mock Test 5 · preserved Physics module

Q1 D	Q2 D	Q3 C	Q4 C	Q5 B
Q6 D	Q7 D	Q8 C	Q9 D	Q10 C
Q11 E	Q12 D	Q13 C	Q14 C	Q15 D
Q16 D	Q17 E	Q18 D	Q19 C	Q20 C
Q21 B	Q22 B	Q23 D	Q24 D	Q25 B
Q26 C	Q27 C			

Biology

Preserved Biology Harder than ESAT source set

Q1 C	Q2 F	Q3 B	Q4 C	Q5 C
Q6 D	Q7 C	Q8 D	Q9 C	Q10 B
Q11 B	Q12 E	Q13 F	Q14 E	Q15 C
Q16 D	Q17 B	Q18 B	Q19 E	Q20 E
Q21 A	Q22 E	Q23 D	Q24 F	Q25 E
Q26 C	Q27 H			

Mathematics 1

27 QUESTIONS • COMPLETE SOLUTIONS

Engineering Mock Test 5 · preserved Mathematics 1 module. Question wording, option order, answer and solution method are unchanged.

M1-01

Laws of indices and exact simplification

Answer C**QUESTION 1**

Simplify fully

$$\frac{(50x^{3/2}y^{-1})^2(2x^{-1}y^{5/2})^3}{(20x^{1/2}y)^2},$$

where x and y are positive.

A $\frac{10y^3\sqrt{y}}{x}$

B $\frac{50y^3}{x}$

C $\frac{50y^3\sqrt{y}}{x}$

D $\frac{50\sqrt{y}}{x}$

E $\frac{250y^3\sqrt{y}}{x}$

WORKED SOLUTION

$$(50x^{3/2}y^{-1})^2 = 2500x^3y^{-2} \text{ and } (2x^{-1}y^{5/2})^3 = 8x^{-3}y^{15/2}.$$

So the numerator is $2500x^3y^{-2} \cdot 8x^{-3}y^{15/2} = 20000y^{11/2}$.

The denominator is $(20x^{1/2}y)^2 = 400xy^2$.

Therefore

$$\frac{20000y^{11/2}}{400xy^2} = \frac{50y^{7/2}}{x} = \boxed{\frac{50y^3\sqrt{y}}{x}}.$$

M1-02

Successive percentage change and comparison

Answer B

QUESTION 2

The price of item P is reduced by 20%, and then a sales tax of 5% is applied to the reduced price.

The price of item Q is increased by 10%, and then reduced by 10%.

After these two-step changes, the final prices of P and Q are equal.

What was the ratio of the original price of P to the original price of Q?

A 28 : 33

B 33 : 28

C 11 : 10

D 6 : 5

E 99 : 80

WORKED SOLUTION

Final price of P is $0.8 \times 1.05P = 0.84P$.

Final price of Q is $1.1 \times 0.9Q = 0.99Q$.

So $0.84P = 0.99Q$, giving

$$\frac{P}{Q} = \frac{0.99}{0.84} = \frac{99}{84} = \frac{33}{28}.$$

M1-03

Algebraic fractions and hidden substitution

Answer D

QUESTION 3

For real x , define

$$E = \frac{\frac{1}{x-3} - \frac{1}{x+2}}{\frac{1}{x-3} + \frac{1}{x+2}}.$$

Given that $x > 3$ and

$$E + \frac{1}{E} = \frac{13}{2},$$

what is the value of x ?

A $\frac{11}{2}$

B 8

C 12

D $\frac{31}{2}$

E 18

WORKED SOLUTION

Simplify first:

$$E = \frac{\frac{(x+2)-(x-3)}{(x-3)(x+2)}}{\frac{(x+2)+(x-3)}{(x-3)(x+2)}} = \frac{5}{2x-1}.$$

Let $t = E$. Then $t + \frac{1}{t} = \frac{13}{2}$, so

$$2t^2 - 13t + 2 = 0 = (2t - 1)(t - 6).$$

Hence $t = 6$ or $t = \frac{1}{6}$. Since $x > 3$, we have $0 < E < 1$, so $E = \frac{1}{6}$.

Therefore

$$\frac{5}{2x-1} = \frac{1}{6} \Rightarrow 2x - 1 = 30 \Rightarrow \boxed{x = \frac{31}{2}}.$$

M1-04

Inequalities with domain restrictions

Answer A

QUESTION 4

How many integer values of x satisfy all three conditions?

- $-5 \leq x \leq 5$
- $\frac{x^2 + 3x - 10}{x^2 - 1}$ is defined
- $\frac{x^2 + 3x - 10}{x^2 - 1} \leq 1$

A 6

B 5

C 4

D 7

WORKED SOLUTION

Bring everything to one side:

$$\frac{x^2 + 3x - 10}{x^2 - 1} - 1 \leq 0 \Rightarrow \frac{3x - 9}{(x - 1)(x + 1)} \leq 0.$$

So we solve

$$\frac{x - 3}{(x - 1)(x + 1)} \leq 0, \quad x \neq \pm 1.$$

A sign chart gives $(-\infty, -1) \cup (1, 3]$.

Intersecting with $-5 \leq x \leq 5$ gives $-5, -4, -3, -2, 2, 3$, so there are 6 values.

M1-05

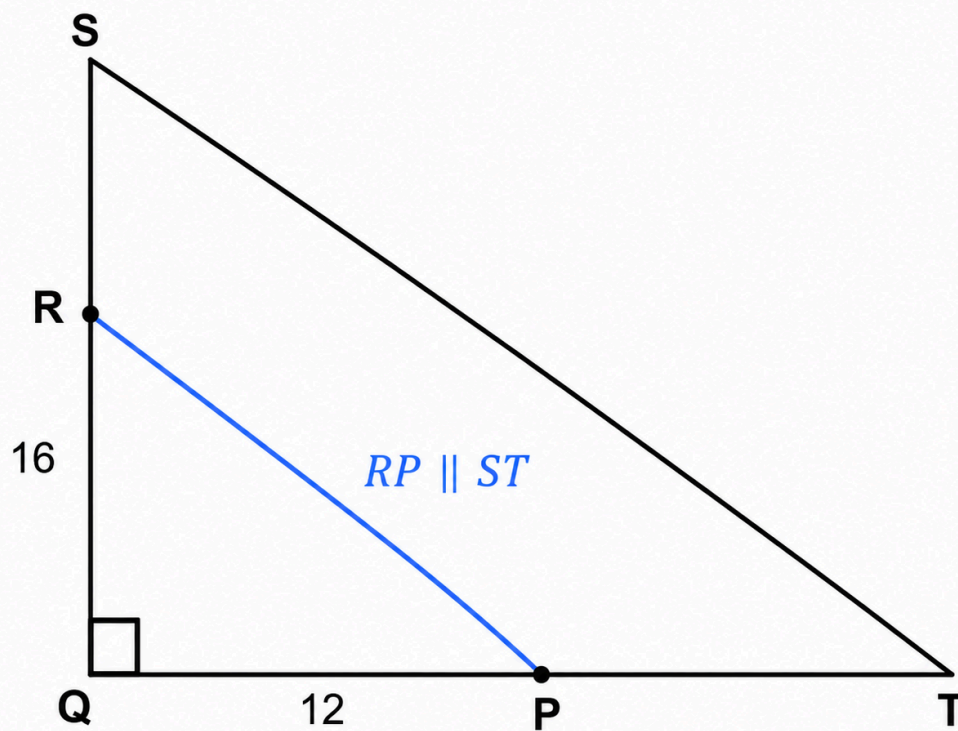
Similarity and area scaling in a right triangle

Answer E

QUESTION 5

Triangle SQT is right-angled at Q, with SQ = 16 cm and QT = 12 cm.

Point R lies on SQ such that SR : RQ = 1 : 3. Through R, a line parallel to ST meets QT at P.



What is the area of quadrilateral SRP T?

A 24 cm^2

B 30 cm^2

C 36 cm^2

D 40 cm^2

E 42 cm^2

WORKED SOLUTION

The area of the whole triangle is $\frac{1}{2} \cdot 16 \cdot 12 = 96$.

Since $SR : RQ = 1 : 3$, we get $RQ = \frac{3}{4} \cdot 16 = 12$.

Because $RP \parallel ST$, triangles QRP and QST are similar with scale factor

$$\frac{QR}{QS} = \frac{12}{16} = \frac{3}{4}.$$

So $QP = \frac{3}{4} \cdot 12 = 9$. Then area of triangle QRP is $\frac{1}{2} \cdot 12 \cdot 9 = 54$.

Hence quadrilateral $SRPT$ has area $96 - 54 = \boxed{42 \text{ cm}^2}$.

M1-06

Arithmetic sequences and common terms

Answer **C**

QUESTION 6

The n th term of sequence T is $4n - 3$.

The n th term of sequence V is $7n + p$, where p is an integer and $0 \leq p < 7$.

The sequences have consecutive common terms 53 and 81.

What is the value of p ?

A 1

B 2

C 4

D 5

E 6

WORKED SOLUTION

Using the common term 53,

$$53 = 7n + p.$$

Since $53 \equiv 4 \pmod{7}$, and $0 \leq p < 7$, we must have $p = 4$.

M1-07

Cylinders with swapped dimensions

Answer **B**

QUESTION 7

Two solid cylinders P and Q are defined using positive numbers x and y , where $x > y$.

- Cylinder P has diameter x and height y .
- Cylinder Q has diameter y and height x .

The total surface area of P is $\frac{8}{5}$ of the total surface area of Q.

The volume of P exceeds the volume of Q by $32\pi \text{ cm}^3$.

What is the value of $x + y$?

A 10

B 12

C 14

D 16

E 18

WORKED SOLUTION

For a cylinder of diameter d and height h , total surface area is

$$2\pi\left(\frac{d}{2}\right)^2 + 2\pi\left(\frac{d}{2}\right)h.$$

Thus

$$A_P = \pi \left(\frac{x^2}{2} + xy \right), \quad A_Q = \pi \left(\frac{y^2}{2} + xy \right).$$

Set $x = ky$. Then

$$\frac{k^2 + 2k}{1 + 2k} = \frac{8}{5}.$$

So $5k^2 - 6k - 8 = 0$, giving the positive root $k = 2$. Hence $x = 2y$.

Now

$$V_P - V_Q = \frac{\pi}{4}(x^2y - y^2x) = \frac{\pi}{4}xy(x - y) = 32\pi.$$

Substitute $x = 2y$:

$$\frac{\pi}{4}(2y)(y)(y) = 32\pi \Rightarrow \frac{y^3}{2} = 32 \Rightarrow y^3 = 64 \Rightarrow y = 4.$$

Hence $x = 8$, so $x + y = \boxed{12}$.

M1-08

Ratio change after equal spending

Answer **D**

QUESTION 8

Pat and Alex have a combined total of £ 84. The ratio of Pat's money to Alex's money is 5 : 2.

They each spend the same amount on sweets.

After this, Pat receives £ 6 and Alex receives £ 1. The ratio of their new amounts is then 3 : 1.

How much did each of them spend?

A £ 3

B £ 4

C £ 4.25

D £ 4.50

E £ 6

WORKED SOLUTION

Initially the amounts are 60 and 24.

If each spends s , then afterwards they have $60 - s$ and $24 - s$. After receiving the extra money, they have $66 - s$ and $25 - s$.

So

$$66 - s = 3(25 - s) \Rightarrow 66 - s = 75 - 3s \Rightarrow 2s = 9 \Rightarrow s = 4.5.$$

Each spent £ 4.50.

M1-09

Coordinate geometry of a tangent

Answer A

QUESTION 9

The circle $x^2 + y^2 = 25$ has centre at the origin.

A tangent to the circle passes through the point $(1, 7)$ and has positive gradient.

What is the equation of the tangent?

A $3x - 4y + 25 = 0$

B $4x - 3y + 25 = 0$

C $3x + 4y - 25 = 0$

D $4x + 3y - 25 = 0$

E

$$y = \frac{4}{3}x + \frac{17}{3}$$

WORKED SOLUTION

Let the tangent be $y = mx + c$. Since it passes through $(1, 7)$, we have $c = 7 - m$.

The distance from the origin to the tangent must equal the radius 5:

$$\frac{|c|}{\sqrt{m^2 + 1}} = 5.$$

So

$$(7 - m)^2 = 25(m^2 + 1) \Rightarrow 12m^2 + 7m - 12 = 0 = (4m - 3)(3m + 4).$$

The positive gradient is $m = \frac{3}{4}$. Then $c = 7 - \frac{3}{4} = \frac{25}{4}$.

Hence

$$y = \frac{3}{4}x + \frac{25}{4} \Leftrightarrow \boxed{3x - 4y + 25 = 0}.$$

M1-10

Conditional probability with non-standard dice

Answer E**QUESTION 10**

Two fair dice X and Y are available.

- The faces on X are labelled 1, 1, 2, 3, 4, 5.
- The faces on Y are labelled 2, 3, 3, 4, 5, 6.

One of the dice is chosen at random and rolled twice.

Given that the total shown is 7, what is the probability that the chosen die was X ?

A

$$\frac{1}{3}$$

B $\frac{3}{8}$

C $\frac{1}{2}$

D $\frac{3}{5}$

E $\frac{2}{5}$

WORKED SOLUTION

For die X, the ordered outcomes summing to 7 are (2, 5), (5, 2), (3, 4), (4, 3): 4 outcomes out of 36. So $P(7 | X) = \frac{1}{9}$.

For die Y, 3 appears twice, so the ordered outcomes summing to 7 are 6 in total. Hence $P(7 | Y) = \frac{1}{6}$.

With equal prior probability of choosing each die,

$$P(X | 7) = \frac{\frac{1}{2} \cdot \frac{1}{9}}{\frac{1}{2} \cdot \frac{1}{9} + \frac{1}{2} \cdot \frac{1}{6}} = \frac{\frac{1}{9}}{\frac{1}{9} + \frac{1}{6}} = \frac{2}{5}.$$

M1-11

Exponential equation via substitution

Answer **C**

QUESTION 11

The equation $2^x + 2^{2-x} = 5$ has two real solutions.

What is the larger possible value of x?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

Let $t = 2^x$. Then $2^{2-x} = \frac{4}{t}$, so

$$t + \frac{4}{t} = 5 \Rightarrow t^2 - 5t + 4 = 0 = (t - 1)(t - 4).$$

Thus $t = 1$ or $t = 4$, so $x = 0$ or $x = 2$. The larger value is 2.

M1-12

Rearranging to make a variable the subject

Answer B

QUESTION 12

Suppose x and y satisfy

$$x = y + \frac{4}{y},$$

where $y > 0$ and $x > 4$.

Which of the following is the larger possible value of y in terms of x ?

A $\frac{x - \sqrt{x^2 - 16}}{2}$

B $\frac{x + \sqrt{x^2 - 16}}{2}$

C $\frac{x + \sqrt{x^2 + 16}}{2}$

D $\frac{2}{x + \sqrt{x^2 - 16}}$

E $\frac{4}{x + \sqrt{x^2 - 16}}$

WORKED SOLUTION

Multiply by y :

$$xy = y^2 + 4 \Rightarrow y^2 - xy + 4 = 0.$$

Using the quadratic formula,

$$y = \frac{x \pm \sqrt{x^2 - 16}}{2}.$$

The larger possible value uses the positive sign, so

$$y = \frac{x + \sqrt{x^2 - 16}}{2}.$$

M1-13

Surface area comparison of a cylinder and a cube

Answer D

QUESTION 13

A solid cylinder has radius r cm and height h cm.

A cube has side length $\sqrt{\pi} r$ cm.

The total surface area of the cylinder is equal to four times the total surface area of the cube.

What is $\frac{h}{r}$?

A 7

B 9

C 10

D 11

E 12

WORKED SOLUTION

The cylinder's total surface area is $2\pi r^2 + 2\pi rh = 2\pi r(r + h)$.

The cube's total surface area is $6(\sqrt{\pi}r)^2 = 6\pi r^2$.

Given the cylinder equals four times the cube,

$$2\pi r(r + h) = 24\pi r^2.$$

Cancel $2\pi r$:

$$r + h = 12r \Rightarrow h = 11r.$$

Therefore $\frac{h}{r} = \boxed{11}$.

M1-14

Mean and median with an ordered list

Answer A

QUESTION 14

Consider the list

$$1, 2, x, x + 1, 2x - 1, 15,$$

where $2 < x < 8$.

The mean of the six numbers is one more than the median.

What is the value of x ?

A $\frac{9}{2}$

B 4

C 5

D $\frac{11}{2}$

E 6

WORKED SOLUTION

Since $x > 2$, the list is already ordered. So the median is

$$\frac{x + (x + 1)}{2} = x + \frac{1}{2}.$$

The mean is one more than this, so the mean is $x + \frac{3}{2}$.

The mean is also

$$\frac{1 + 2 + x + (x + 1) + (2x - 1) + 15}{6} = \frac{4x + 18}{6}.$$

Therefore

$$\frac{4x + 18}{6} = x + \frac{3}{2} \Rightarrow 4x + 18 = 6x + 9 \Rightarrow 2x = 9 \Rightarrow \boxed{x = \frac{9}{2}}.$$

QUESTION 15

A cyclist rides from A to B uphill at 12 km h^{-1} , immediately turns round, and returns from B to A along the same route at 18 km h^{-1} .

She spends exactly 6 minutes stationary at B.

The total time for the complete trip is 46 minutes.

What is the distance from A to B?

A 3.6 km

B 4.0 km

C 4.2 km

D 4.5 km

E 4.8 km

WORKED SOLUTION

Excluding the 6-minute stop, the travel time is 40 minutes = $\frac{2}{3}$ hours.

If the one-way distance is d km, then

$$\frac{d}{12} + \frac{d}{18} = \frac{2}{3}.$$

Since $\frac{1}{12} + \frac{1}{18} = \frac{5}{36},$

$$d \cdot \frac{5}{36} = \frac{2}{3} \Rightarrow d = \frac{2}{3} \cdot \frac{36}{5} = \frac{24}{5} = 4.8.$$

So the distance is 4.8 km.

M1-16

Exact trigonometric values

Answer **B**

QUESTION 16

Given that

$$y = \frac{2 \sin 15^\circ}{\cos 30^\circ},$$

what is the value of y^2 ?

A $\frac{4 - 2\sqrt{3}}{3}$

B $\frac{8 - 4\sqrt{3}}{3}$

C $\frac{8 + 4\sqrt{3}}{3}$

D $2 - \sqrt{3}$

E $4 - 2\sqrt{3}$

WORKED SOLUTION

Use

$$\sin 15^\circ = \sin(45^\circ - 30^\circ) = \frac{\sqrt{6} - \sqrt{2}}{4}, \quad \cos 30^\circ = \frac{\sqrt{3}}{2}.$$

Hence

$$y = \frac{2 \cdot \frac{\sqrt{6} - \sqrt{2}}{4}}{\frac{\sqrt{3}}{2}} = \frac{\sqrt{6} - \sqrt{2}}{\sqrt{3}}.$$

Therefore

$$y^2 = \frac{(\sqrt{6} - \sqrt{2})^2}{3} = \frac{6 + 2 - 2\sqrt{12}}{3} = \frac{8 - 4\sqrt{3}}{3}.$$

M1-17

Inverse proportion in standard form

Answer C

QUESTION 17

The variable t is inversely proportional to the square of w .

When $w = 2 \times 10^3$, the value of t is 4.5×10^{-5} .

What is the value of w when $t = 8 \times 10^{-6}$?

A 7.5×10^3

B $3\sqrt{10} \times 10^2$

C $1.5\sqrt{10} \times 10^3$

D $2\sqrt{10} \times 10^3$

E 4.5×10^3

WORKED SOLUTION

Since $t = \frac{k}{w^2}$, use the first condition:

$$k = (4.5 \times 10^{-5})(2 \times 10^3)^2 = (4.5 \times 10^{-5})(4 \times 10^6) = 180.$$

Now solve

$$8 \times 10^{-6} = \frac{180}{w^2} \Rightarrow w^2 = 2.25 \times 10^7.$$

So

$$w = \sqrt{2.25 \times 10^7} = 1.5\sqrt{10} \times 10^3.$$

M1-18

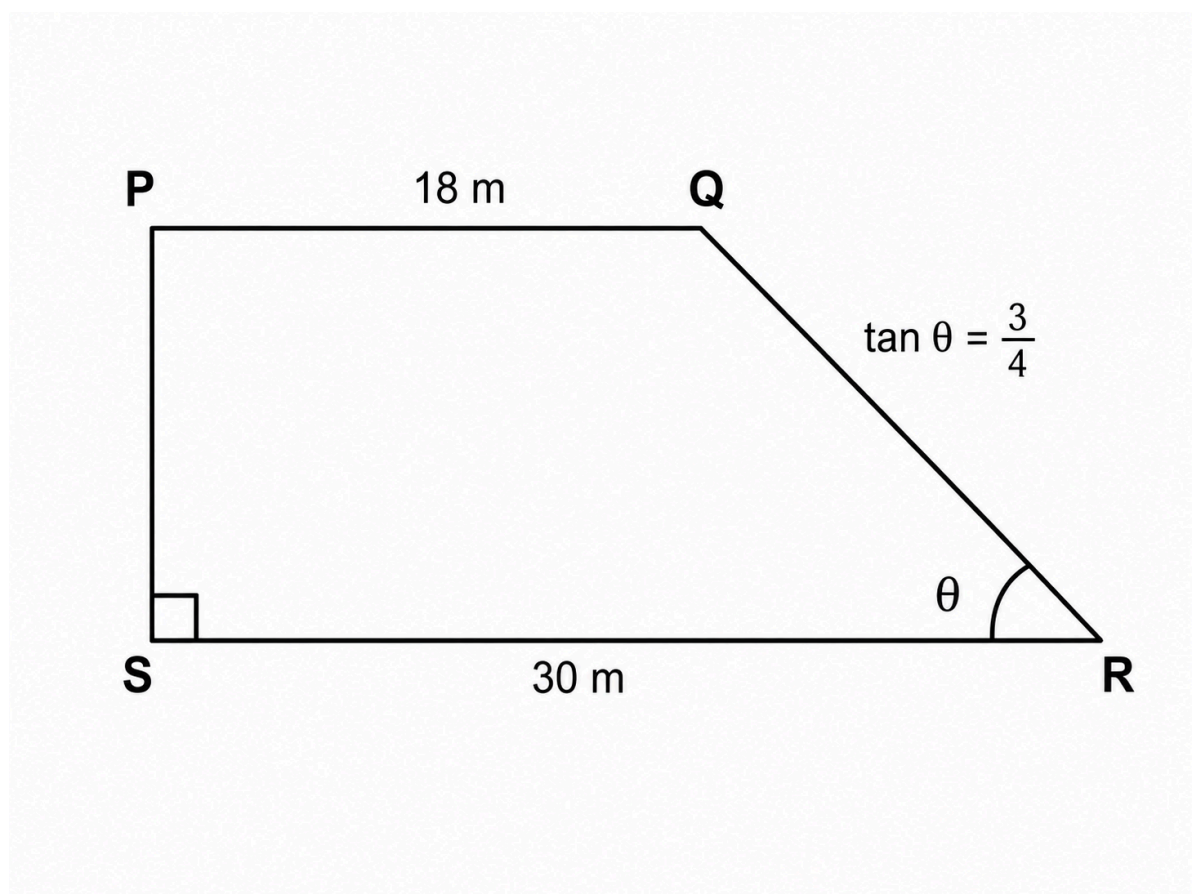
Trapezium area using trigonometry

Answer D

QUESTION 18

PQRS is a trapezium in which $PQ \parallel SR$ and $PS \perp SR$.

Also, $PQ = 18 \text{ m}$, $SR = 30 \text{ m}$, and $\tan \angle QRS = \frac{3}{4}$.



What is the area of the trapezium?

A 180 m^2

B 192 m^2

C 204 m^2

D 216 m^2

E 240 m^2

WORKED SOLUTION

The difference between the parallel sides is $30 - 18 = 12$, which is the horizontal run of the sloping side.

If the height is h , then

$$\tan \angle QRS = \frac{h}{12} = \frac{3}{4} \Rightarrow h = 9.$$

So the area is

$$\frac{1}{2}(18 + 30)(9) = 24 \cdot 9 = \boxed{216 \text{ m}^2}.$$

M1-19

Quadratic roots and discriminant reasoning

Answer A

QUESTION 19

The equation $x^2 + bx + 8 = 0$ has two real roots whose positive difference is 2. The sum of the roots is negative.

What is the value of b ?

A 6

B 5

C 4

D -6

E -4

WORKED SOLUTION

If the roots are r and s , then $(r - s)^2 = (r + s)^2 - 4rs$.

Here $r + s = -b$ and $rs = 8$. Since the difference is 2,

$$4 = b^2 - 32 \Rightarrow b^2 = 36.$$

So $b = \pm 6$. The sum of the roots is negative, so $-b < 0 \Rightarrow b > 0$. Hence

$$\boxed{b = 6}.$$

M1-20

Algebraic geometry with area change

Answer **E**

QUESTION 20

A rectangle has length $(2x - 1)$ cm and width $(x + 1)$ cm.

A larger rectangle is made by adding 3 cm to both the length and the width.

The larger rectangle has twice the area of the original rectangle.

What is the perimeter of the original rectangle?

A 24 cm

B 26 cm

C 28 cm

D 29 cm

E 30 cm

WORKED SOLUTION

Original area is $(2x - 1)(x + 1)$. New dimensions are $2x + 2$ and $x + 4$, so the new area is $(2x + 2)(x + 4)$.

Given new area is twice old area,

$$(2x + 2)(x + 4) = 2(2x - 1)(x + 1).$$

This simplifies to

$$x^2 - 4x - 5 = 0 = (x - 5)(x + 1).$$

So $x = 5$. The original dimensions are 9 and 6, so the perimeter is

$$2(9 + 6) = \boxed{30 \text{ cm}}.$$

M1-21

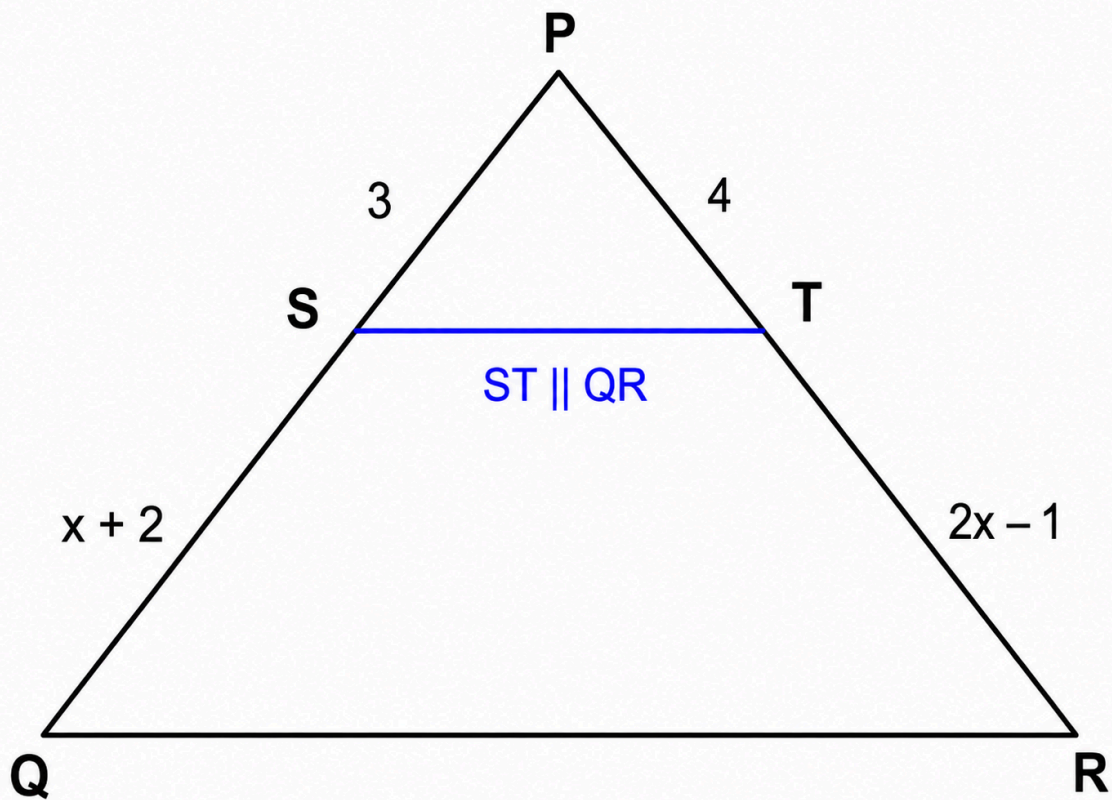
Similarity and area scaling

Answer **C**

QUESTION 21

In triangle PQR , point S lies on PQ and point T lies on PR . The line ST is parallel to QR .

The lengths are labelled as follows: $PS = 3 \text{ cm}$, $SQ = (x + 2) \text{ cm}$, $PT = 4 \text{ cm}$, $TR = (2x - 1) \text{ cm}$.



Given that the area of triangle PST is 12 cm^2 , what is the area of trapezium $STQR$?

A 96 cm^2

B 120 cm^2

C 135 cm^2

D 144 cm^2

E 147 cm^2

WORKED SOLUTION

Since $ST \parallel QR$, triangles PST and PQR are similar, so

$$\frac{PS}{PQ} = \frac{PT}{PR}.$$

Now $PQ = 3 + (x + 2) = x + 5$ and $PR = 4 + (2x - 1) = 2x + 3$.

Therefore

$$\frac{3}{x + 5} = \frac{4}{2x + 3}.$$

So $6x + 9 = 4x + 20$, giving $x = \frac{11}{2}$.

Then

$$\frac{PS}{PQ} = \frac{3}{21/2} = \frac{2}{7}.$$

Hence the area scale factor is $\left(\frac{2}{7}\right)^2 = \frac{4}{49}$.

So if area $PS T = 12$, then area $PQR = 12 \cdot \frac{49}{4} = 147$.

Thus the area of trapezium $STQR$ is $147 - 12 = \boxed{135 \text{ cm}^2}$.

M1-22

Probability with two colours

Answer **B**

QUESTION 22

Charlie has a bowl containing red sweets and green sweets only. There are 9 more red sweets than green sweets.

Two sweets are chosen at random without replacement.

The probability that the two sweets are the same colour is $\frac{4}{7}$.

How many sweets are there in the bowl altogether?

A 18

B 21

C 24

D 27

E 30

WORKED SOLUTION

Let the number of green sweets be g . Then the number of red sweets is $g + 9$, so total sweets are $2g + 9$.

The probability condition is

$$\frac{\binom{g+9}{2} + \binom{g}{2}}{\binom{2g+9}{2}} = \frac{4}{7}.$$

This simplifies to

$$(g - 6)(g + 18) = 0.$$

So $g = 6$. Then red sweets = 15, giving total sweets = 21.

M1-23

Successive translations and vector equations

Answer **D**

QUESTION 23

The point $(-1, 5)$ is translated by two successive vectors.

The first translation is by $\begin{pmatrix} 3p \\ 2 - q \end{pmatrix}$, and the second is by $\begin{pmatrix} q + 1 \\ p - 3 \end{pmatrix}$.

The final image is $(8, 4)$.

What is the value of $p^2 + q^2$?

A 4

B 5

C 6

D 8

E 10

WORKED SOLUTION

The total translation vector is

$$\begin{pmatrix} 3p + q + 1 \\ p - q - 1 \end{pmatrix}.$$

Since $(-1, 5) \rightarrow (8, 4)$, the net translation is $\begin{pmatrix} 9 \\ -1 \end{pmatrix}$.

So

$$3p + q = 8, \quad p - q = 0.$$

Thus $p = q$, so $4p = 8 \Rightarrow p = 2$, hence $q = 2$.

Therefore $p^2 + q^2 = 4 + 4 = \boxed{8}$.

M1-24

Quadratic graphs and minimum values

Answer **A**

QUESTION 24

Consider the family of graphs

$$y = ax^2 + 2(a - 1)x + a + 3.$$

What is the complete range of values of a for which the minimum point of the graph lies above the x -axis?

A $a > \frac{1}{5}$

B $a > 0$

C

$$a \geq \frac{1}{5}$$

D

$$a < 0$$

E

$$0 < a < \frac{1}{5}$$

WORKED SOLUTION

For there to be a minimum, we need $a > 0$.

For $y = ax^2 + bx + c$, the minimum value is $c - \frac{b^2}{4a}$.

Here $b = 2(a - 1)$ and $c = a + 3$, so the minimum value is

$$a + 3 - \frac{(2(a - 1))^2}{4a} = a + 3 - \frac{(a - 1)^2}{a} = 5 - \frac{1}{a}.$$

We need this to be positive:

$$5 - \frac{1}{a} > 0.$$

Since $a > 0$, this gives $5a > 1 \Rightarrow a > \frac{1}{5}$.

M1-25

Counting constrained selections

Answer **E****QUESTION 25**

An online company sells storage containers. The following items are available:

Capacity of container	Number available
2 litres	2
3 litres	3

Capacity of container	Number available
5 litres	4

A customer chooses a selection of containers with total capacity 23 litres.

Different selections are distinguished only by how many containers of each size are chosen.

If all possible selections are equally likely, what is the probability that the chosen selection contains at least one 2-litre container?

A $\frac{1}{3}$

B $\frac{1}{2}$

C $\frac{3}{5}$

D $\frac{3}{4}$

E $\frac{2}{3}$

WORKED SOLUTION

Let a , b , c be the numbers of 2-, 3- and 5-litre containers. Then

$$2a + 3b + 5c = 23,$$

with $0 \leq a \leq 2$, $0 \leq b \leq 3$, $0 \leq c \leq 4$.

The valid selections are

- (0, 1, 4)
- (1, 2, 3)
- (2, 3, 2)

Two of the three contain at least one 2-litre container, so the probability is $\frac{2}{3}$.

M1-26

Similarity and three-dimensional scaling

Answer C

QUESTION 26

The space diagonal of a cuboid is $7\sqrt{13}$ cm.

A similar cuboid has volume 8 times the volume of the original cuboid.

What is the space diagonal of the larger cuboid?

A $7\sqrt{26}$ cm

B $14\sqrt{7}$ cm

C $14\sqrt{13}$ cm

D $21\sqrt{13}$ cm

E $28\sqrt{13}$ cm

WORKED SOLUTION

A volume scale factor of 8 means a linear scale factor of $\sqrt[3]{8} = 2$.

So the new diagonal is

$$2 \cdot 7\sqrt{13} = \boxed{14\sqrt{13} \text{ cm}}.$$

QUESTION 27

Alex, Cameron and Sam are running a 400 m race. They each run at a different constant speed.

When Alex finishes, Cameron has run 300 m and Sam has run 240 m.

Alex beats Sam by $33\frac{1}{3}$ seconds.

What is Cameron's speed?

A 5 m s^{-1}

B 6 m s^{-1}

C 6.4 m s^{-1}

D 7 m s^{-1}

E 8 m s^{-1}

WORKED SOLUTION

Let Alex's speed be A , Cameron's speed be C , and Sam's speed be S .

When Alex finishes, the elapsed time is $\frac{400}{A}$. In that time, Cameron runs 300 m, so

$$C \cdot \frac{400}{A} = 300 \Rightarrow \frac{C}{A} = \frac{3}{4}.$$

Similarly, Sam runs 240 m, so

$$S \cdot \frac{400}{A} = 240 \Rightarrow \frac{S}{A} = \frac{3}{5}.$$

$$\text{Thus } S = \frac{3}{5}A.$$

Alex beats Sam by $\frac{100}{3}$ s, so

$$\frac{400}{S} - \frac{400}{A} = \frac{100}{3}.$$

Substitute $S = \frac{3}{5}A$:

$$\frac{400}{3A/5} - \frac{400}{A} = \frac{100}{3} \Rightarrow \frac{2000}{3A} - \frac{400}{A} = \frac{100}{3} \Rightarrow \frac{800}{3A} = \frac{100}{3}.$$

So $A = 8 \text{ m s}^{-1}$. Hence

$$C = \frac{3}{4}A = 6 \text{ m s}^{-1}.$$

Therefore Cameron's speed is 6 m s^{-1} .

Physics

27 QUESTIONS • COMPLETE SOLUTIONS

Engineering Mock Test 5 · preserved Physics module. Question wording, option order, answer and solution method are unchanged.

P-01

Charge, current, power and heating

Answer D**QUESTION 1**

A resistor has a constant potential difference of 9.0 V across it. A total charge of 180 C passes through it in 4.00 min .

The resistor is then left connected for a total of 6.00 min . Assume all the energy dissipated warms a 0.250 kg metal block of specific heat capacity $400 \text{ J kg}^{-1}\text{K}^{-1}$, and that there are no losses.

What is the temperature rise of the block?

A 12.0 K **B** 18.0 K **C** 21.6 K **D** 24.3 K **E** 27.0 K **WORKED SOLUTION**

The current is

$$I = \frac{Q}{t} = \frac{180}{240} = 0.75 \text{ A.}$$

So the power is

$$P = VI = 9.0 \times 0.75 = 6.75 \text{ W}.$$

Over 6.00 min = 360 s, the energy transferred is

$$E = Pt = 6.75 \times 360 = 2430 \text{ J}.$$

Hence

$$\Delta T = \frac{E}{mc} = \frac{2430}{0.250 \times 400} = 24.3 \text{ K}.$$

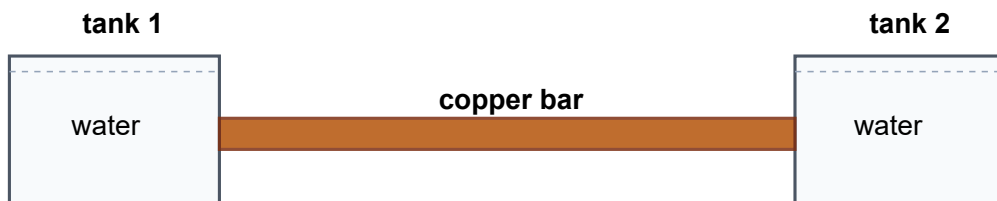
P-02

Thermal conduction and scaling

Answer D

QUESTION 2

Two large copper tanks of water are connected by a solid cylindrical copper bar. Initially the water temperatures are 20°C and 80°C , and the bar has length L and diameter d .



The system is modified so that the temperatures become 35°C and 95°C , the bar length becomes $0.75L$, and the diameter becomes $1.5d$.

By what factor does the rate of conduction of thermal energy along the bar change?

A 0.75

B 1.5

C 2.25

D 3.0

E 4.0

WORKED SOLUTION

For conduction,

$$P \propto \frac{A \Delta T}{L}.$$

The temperature difference stays 60°C , so ΔT is unchanged. The area factor is

$$\left(\frac{1.5d}{d}\right)^2 = 2.25.$$

The length factor is

$$\frac{L}{0.75L} = \frac{4}{3}.$$

Hence the overall factor is

$$2.25 \times \frac{4}{3} = 3.0.$$

P-03

Forces, towing and tension

Answer **C**

QUESTION 3

A car of mass 1400 kg tows a caravan of mass 1000 kg along a horizontal road at constant speed. The driving force from the car's engine is then increased by 3000 N.

At one instant during the resulting acceleration, the resistive force on the car has increased by 200 N and the resistive force on the caravan has increased by 400 N, compared with their values at constant speed.

By how much has the tension in the tow bar increased compared with its value at constant speed?

A 600 N

B 1000 N

C 1400 N

D 1600 N

E 1800 N

WORKED SOLUTION

For the combined car–caravan system, the increase in the forward resultant is

$$3000 - 200 - 400 = 2400 \text{ N.}$$

The total mass is 2400 kg, so the acceleration is

$$a = \frac{2400}{2400} = 1.0 \text{ m s}^{-2}.$$

Let T_0 be the original tow-bar tension and R_0 the original resistive force on the caravan. At constant speed, $T_0 = R_0$.

During the acceleration, let the new tension be T_1 . The caravan equation is

$$T_1 - (R_0 + 400) = 1000(1.0).$$

Hence $T_1 - R_0 = 1400 \text{ N}$. Since $T_0 = R_0$,

$$T_1 - T_0 = \boxed{1400 \text{ N}}.$$

QUESTION 4

A small piece of space debris of mass 0.10 kg strikes the International Space Station at a relative speed of $1.5 \times 10^4 \text{ m s}^{-1}$. The debris comes to rest relative to the station in 0.010 s .

Assume the average contact area during the impact is 2.0 cm^2 .

What is the average pressure exerted on the station during the impact?

A $7.5 \times 10^6 \text{ Pa}$

B $7.5 \times 10^7 \text{ Pa}$

C $7.5 \times 10^8 \text{ Pa}$

D $1.5 \times 10^8 \text{ Pa}$

E $1.5 \times 10^9 \text{ Pa}$

WORKED SOLUTION

The average force magnitude is

$$F = \frac{\Delta p}{\Delta t} = \frac{0.10 \times 1.5 \times 10^4}{0.010} = 1.5 \times 10^5 \text{ N}.$$

The contact area is

$$2.0 \text{ cm}^2 = 2.0 \times 10^{-4} \text{ m}^2.$$

So the pressure is

$$p = \frac{F}{A} = \frac{1.5 \times 10^5}{2.0 \times 10^{-4}} = 7.5 \times 10^8 \text{ Pa}.$$

QUESTION 5

A star is moving away from a space telescope. The star emits light of wavelength 500 nm, but the telescope detects it with wavelength 502 nm.

What is the percentage decrease in the detected frequency, and hence in the energy of each detected photon?

A 0.20%

B 0.40%

C 0.80%

D 1.0%

E 2.0%

WORKED SOLUTION

In vacuum, $c = f\lambda$, so $f \propto \frac{1}{\lambda}$.

The fractional decrease in frequency is therefore approximately the same as the fractional increase in wavelength:

$$\frac{502 - 500}{502} \approx \frac{2}{502} \approx 0.00398.$$

This is about 0.40%. Since photon energy is $E = hf$, the percentage decrease in photon energy is the same.

QUESTION 6

An electric motor is supplied with power at 0.800 kW. Its efficiency is 60%, and it runs for half an hour while lifting a load vertically at constant speed.

Ignoring any change in the motor itself, what maximum mass can be lifted through a height of 180 m? Take $g = 10 \text{ N kg}^{-1}$.

A 120 kg

B 240 kg

C 360 kg

D 480 kg

E 720 kg

WORKED SOLUTION

Input power is 800 W, so useful power is

$$0.60 \times 800 = 480 \text{ W.}$$

Over 1800 s, the useful energy is

$$E = 480 \times 1800 = 864000 \text{ J.}$$

This equals the gain in gravitational potential energy:

$$mgh = 864000.$$

Hence

$$m = \frac{864000}{10 \times 180} = 480 \text{ kg.}$$

QUESTION 7

Air is trapped in a cylinder by a freely moving piston. Initially the trapped air has density ρ at temperature 300 K.

The external load on the piston is adjusted so that the pressure of the trapped air becomes 20% greater than its initial pressure. The load is then kept unchanged, so the pressure remains at this value while the trapped air cools to 270 K. No gas escapes.

Assume the air behaves as an ideal gas. What is the final density of the trapped air?

A 1.09ρ

B 1.20ρ

C 1.25ρ

D $\frac{4}{3}\rho$

E 1.44ρ

WORKED SOLUTION

For a fixed mass of an ideal gas,

$$p = \rho RT \quad \Rightarrow \quad \rho \propto \frac{p}{T}.$$

The final pressure is 1.20 times the initial pressure, while the temperature changes from 300 K to 270 K. Therefore

$$\frac{\rho_f}{\rho_i} = \frac{1.20}{270/300} = 1.20 \times \frac{300}{270} = \frac{4}{3}.$$

Hence $\rho_f = \frac{4}{3}\rho$.

P-08

Current, magnetic force and geometry

Answer C

QUESTION 8

A steady current flows in a conducting wire. A charge of 48 C passes a point on the wire in 2.0 min.

A straight 36 cm section of the wire lies in a uniform magnetic field of strength 0.20 T. This section makes an angle of 30° to the direction of the field.

What is the magnitude of the magnetic force on this section?

A $2.88 \times 10^{-3} \text{ N}$

B $7.2 \times 10^{-3} \text{ N}$

C $1.44 \times 10^{-2} \text{ N}$

D $2.88 \times 10^{-2} \text{ N}$

E $5.76 \times 10^{-2} \text{ N}$

WORKED SOLUTION

The current is

$$I = \frac{Q}{t} = \frac{48}{120} = 0.40 \text{ A.}$$

The magnetic force on a wire is

$$F = BIL \sin \theta.$$

So

$$F = 0.20 \times 0.40 \times 0.36 \times \sin 30^\circ = 0.20 \times 0.40 \times 0.36 \times 0.5 = 1.44 \times 10^{-2}$$

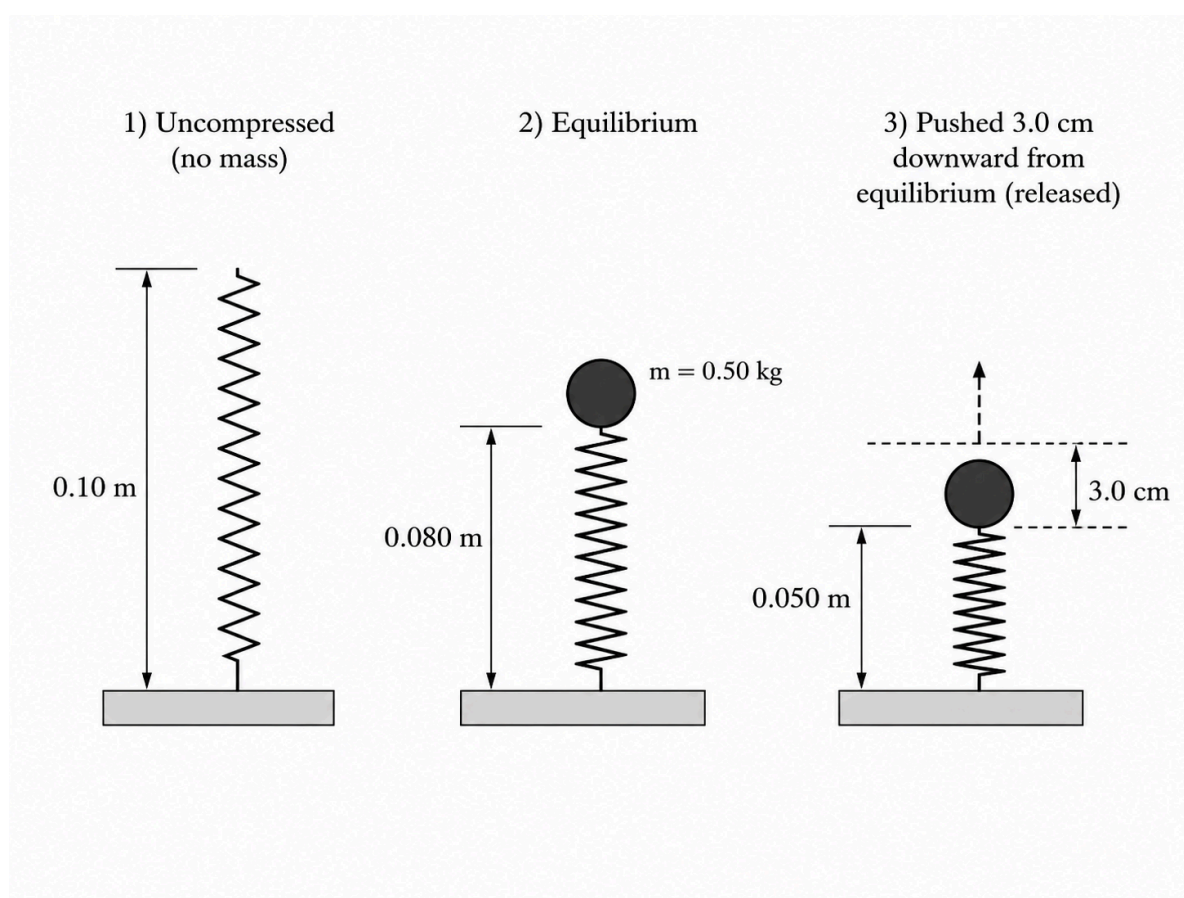
P-09

Springs, equilibrium and oscillation energy

Answer D

QUESTION 9

A light spring has uncompressed length 0.10 m. When an object of mass 0.50 kg rests on top of the spring in equilibrium, the spring length is 0.080 m.



The mass is then pushed a further 3.0 cm downward from equilibrium and released, moving on a frictionless vertical guide.

What is the speed of the mass as it passes back through its equilibrium position?

Take $g = 10 \text{ N kg}^{-1}$.

A 0.30 m s^{-1}

B 0.47 m s^{-1}

C 0.60 m s^{-1}

D 0.67 m s^{-1}

E 0.95 m s^{-1}

WORKED SOLUTION

At equilibrium, the spring is compressed by 0.020 m , so

$$kx = mg \Rightarrow k = \frac{0.50 \times 10}{0.020} = 250 \text{ N m}^{-1}.$$

If the mass is displaced by an additional $a = 0.030 \text{ m}$ from equilibrium, then at the equilibrium position all of the extra elastic/gravitational energy has become kinetic energy:

$$\frac{1}{2}ka^2 = \frac{1}{2}mv^2.$$

Hence

$$v = a\sqrt{\frac{k}{m}} = 0.030\sqrt{\frac{250}{0.50}} = 0.030\sqrt{500} \approx 0.67 \text{ m s}^{-1}.$$

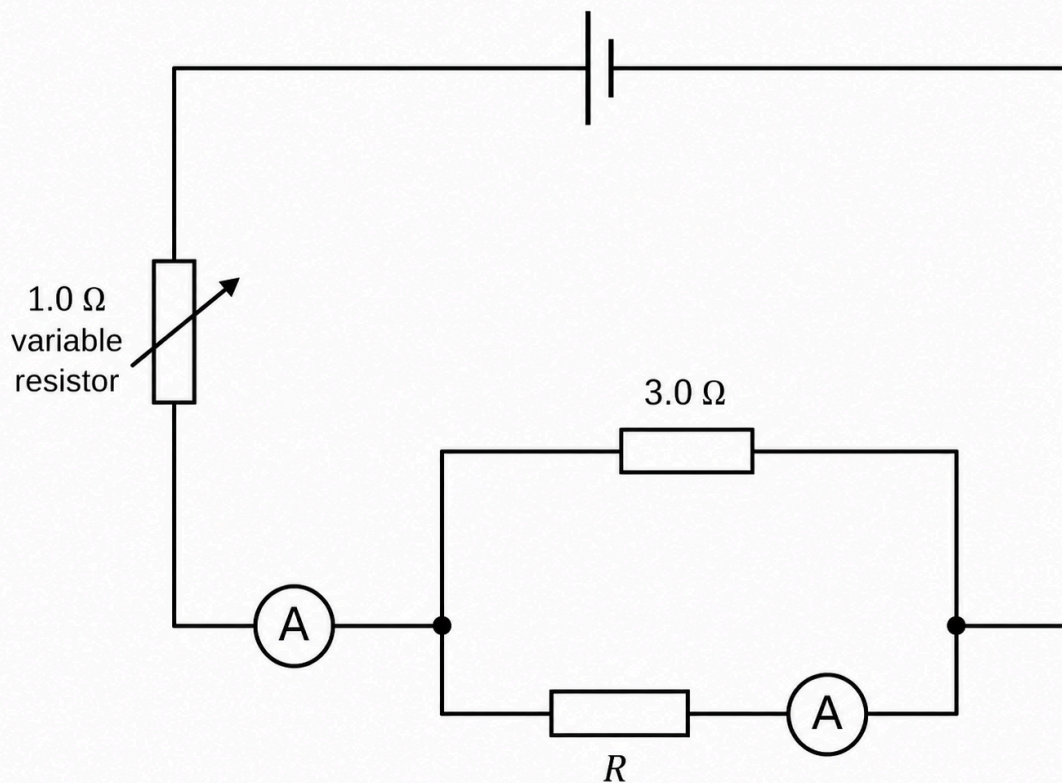
P-10

Circuits, current splitting and emf

Answer **C**

QUESTION 10

The circuit includes a variable resistor in series with a parallel network containing a 3.0Ω resistor and a resistor R . The two ammeter readings are shown.



The variable resistor has resistance $1.0\ \Omega$. Assume ideal ammeters and negligible internal resistance of the supply.

What is the emf of the supply?

A $4.5\ \text{V}$

B $6.6\ \text{V}$

C $7.8\ \text{V}$

D $9.0\ \text{V}$

E $12.3\ \text{V}$

WORKED SOLUTION

The right ammeter is in the R branch, so the current in R is $1.8\ \text{A}$.

The current in the $3.0\ \Omega$ branch is

$$3.3 - 1.8 = 1.5 \text{ A.}$$

So the p.d. across the parallel network is

$$V_{\parallel} = 1.5 \times 3.0 = 4.5 \text{ V.}$$

The current through the series variable resistor is the total current 3.3 A, so its p.d. is

$$V_{\text{var}} = 3.3 \times 1.0 = 3.3 \text{ V.}$$

Hence the supply emf is

$$E = 4.5 + 3.3 = 7.8 \text{ V.}$$

P-11

Nuclides and algebraic reasoning

Answer E

QUESTION 11

Element Q has several isotopes. Consider the two isotopes

$${}_{x}^{3x-7}\text{Q} \quad \text{and} \quad {}_{x}^{2x+3}\text{Q}.$$

The first contains 6 more neutrons than the second.

Another isotope of Q is

$${}_{x}^{3x+1}\text{Q}.$$

How many neutrons does this third isotope contain?

A 19

B 21

C 25

D 29

WORKED SOLUTION

The neutron numbers are:

$$\text{first isotope: } (3x - 7) - x = 2x - 7$$

$$\text{second isotope: } (2x + 3) - x = x + 3$$

Given that the first has 6 more neutrons,

$$(2x - 7) - (x + 3) = 6.$$

So

$$x - 10 = 6 \Rightarrow x = 16.$$

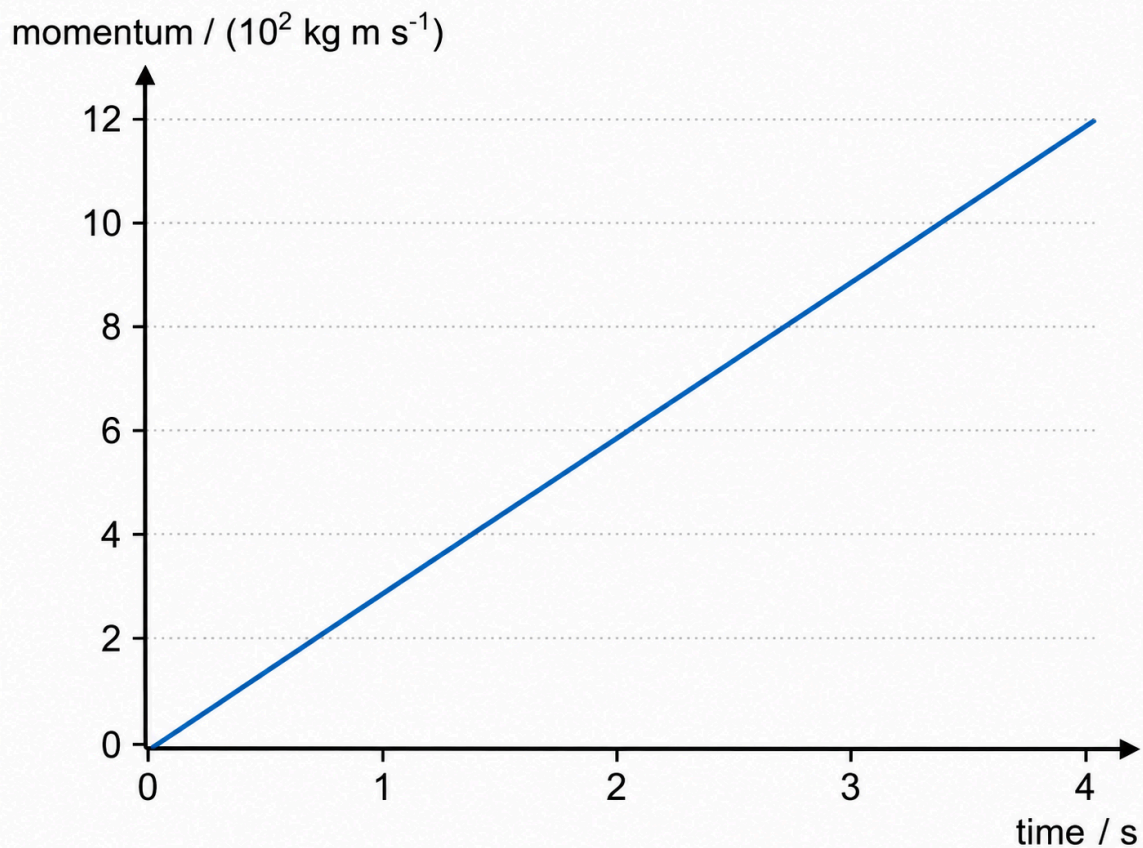
For the third isotope, the number of neutrons is

$$(3x + 1) - x = 2x + 1 = 2(16) + 1 = 33.$$

P-12**Momentum graphs, friction and power****Answer D****QUESTION 12**

A motorised trolley of mass 2.4×10^3 kg accelerates from rest along a horizontal track. The friction force is constant at 1.8×10^3 N.

The graph shows how the momentum of the trolley varies with time.



What is the power delivered by the trolley's motor at $t = 4.0 \text{ s}$?

A $1.5 \times 10^2 \text{ W}$

B $3.0 \times 10^2 \text{ W}$

C $6.3 \times 10^2 \text{ W}$

D $1.05 \times 10^3 \text{ W}$

E $2.1 \times 10^3 \text{ W}$

WORKED SOLUTION

The gradient of a momentum–time graph gives the resultant force. From the graph, the momentum reaches $12 \times 10^2 \text{ kg m s}^{-1}$ at 4.0 s , so

$$F_{\text{net}} = \frac{12 \times 10^2}{4} = 3.0 \times 10^2 \text{ N}.$$

The motor must provide both the resultant force and the force needed to overcome friction:

$$F_{\text{motor}} = 3.0 \times 10^2 + 1.8 \times 10^3 = 2.1 \times 10^3 \text{ N}.$$

At $t = 4.0 \text{ s}$,

$$v = \frac{p}{m} = \frac{1.2 \times 10^3}{2.4 \times 10^3} = 0.50 \text{ m s}^{-1}.$$

Therefore the power delivered is

$$P = F_{\text{motor}}v = (2.1 \times 10^3)(0.50) = \boxed{1.05 \times 10^3 \text{ W}}.$$

P-13

Work–energy and resistive forces

Answer C

QUESTION 13

A block of mass 6.0 kg is pushed along a rough horizontal surface by a constant force of 8.0 N . It accelerates uniformly from rest and reaches speed 2.0 m s^{-1} after 4.0 s .

What fraction of the work done by the pushing force during this 4.0 s is converted into thermal energy against resistive forces?

A $\frac{3}{8}$

B $\frac{1}{2}$

C $\frac{5}{8}$

D $\frac{3}{4}$

WORKED SOLUTION

The acceleration is

$$a = \frac{2.0}{4.0} = 0.50 \text{ m s}^{-2}.$$

The resultant force is therefore

$$F_{\text{net}} = ma = 6.0 \times 0.50 = 3.0 \text{ N}.$$

So the resistive force is

$$8.0 - 3.0 = 5.0 \text{ N}.$$

The displacement in 4.0 s is

$$s = \frac{1}{2}at^2 = \frac{1}{2}(0.50)(4.0)^2 = 4.0 \text{ m}.$$

Work done by the pushing force:

$$W_{\text{push}} = 8.0 \times 4.0 = 32 \text{ J}.$$

Thermal energy against resistance:

$$W_{\text{res}} = 5.0 \times 4.0 = 20 \text{ J}.$$

Hence the fraction is

$$\frac{20}{32} = \frac{5}{8}.$$

QUESTION 14

A metal block of mass M is heated at a constant rate P . The graph of temperature rise ΔT against time t is a straight line of gradient k .

A second block is made of the same material, has mass $3M$, and is heated at constant rate $2P$.

What is the gradient of its ΔT -against- t graph?

A $\frac{k}{6}$

B $\frac{k}{3}$

C $\frac{2k}{3}$

D $\frac{3k}{2}$

E $6k$

WORKED SOLUTION

For the first block,

$$P = Mc \frac{dT}{dt}.$$

Hence

$$k = \frac{P}{Mc}.$$

For the second block, the gradient is

$$k_2 = \frac{2P}{3M_c} = \frac{2}{3} \frac{P}{M_c} = \frac{2k}{3}.$$

P-15

Transformers, efficiency and heating

Answer D

QUESTION 15

A non-ideal transformer has 100 turns on the primary coil and 25 turns on the secondary coil. It is supplied with 3.0 kW of electrical power at a current of 12.5 A.

The output voltage is the same as for an ideal transformer, but the current in the secondary coil is 40 A.

The transformer runs for 15 min. Assume all power lost by inefficiency remains in the transformer, whose thermal capacity is $2.0 \times 10^4 \text{ J K}^{-1}$.

What is the temperature rise of the transformer?

A 12 K

B 18 K

C 24 K

D 27 K

E 30 K

WORKED SOLUTION

Input voltage is

$$V_p = \frac{P_{\text{in}}}{I_p} = \frac{3000}{12.5} = 240 \text{ V}.$$

For the turns ratio,

$$\frac{V_s}{V_p} = \frac{25}{100} = \frac{1}{4},$$

so $V_s = 60 \text{ V}$.

Thus the output power is

$$P_{\text{out}} = V_s I_s = 60 \times 40 = 2400 \text{ W}.$$

Power lost is

$$P_{\text{loss}} = 3000 - 2400 = 600 \text{ W}.$$

In 900 s, the thermal energy gained is

$$E = 600 \times 900 = 5.4 \times 10^5 \text{ J}.$$

So the temperature rise is

$$\Delta T = \frac{E}{C} = \frac{5.4 \times 10^5}{2.0 \times 10^4} = 27 \text{ K}.$$

P-16

Series circuits and power scaling

Answer **D**

QUESTION 16

A set of decorative lights consists of 20 identical lamps connected in series to a d.c. supply of constant voltage. The total power transferred by all 20 lamps together is P .

One lamp fails so that it becomes short-circuited and has zero resistance. The remaining 19 lamps are still lit.

By what factor does the power transferred to *each remaining lamp* increase?

A $\frac{19}{20}$

B $\frac{20}{19}$

C $\frac{361}{400}$

D $\frac{400}{361}$

E $\frac{400}{19}$

WORKED SOLUTION

If each lamp has resistance R , then the original total resistance is $20R$, so current is

$$I = \frac{V}{20R}.$$

After one lamp is short-circuited, the total resistance is $19R$, so current becomes

$$I' = \frac{V}{19R}.$$

Power in each functioning lamp is proportional to $I^2 R$. Therefore the factor increase is

$$\left(\frac{I'}{I}\right)^2 = \left(\frac{V/(19R)}{V/(20R)}\right)^2 = \left(\frac{20}{19}\right)^2 = \frac{400}{361}.$$

P-17

Electromagnetic spectrum

Answer **E**

QUESTION 17

The wavelength range of visible light is about 400 nm to 700 nm. Light of frequency 6.0×10^{14} Hz is green. Microwaves used in cooking have wavelength 12 cm.

Which statements are correct?

1. Light of frequency 7.5×10^{14} Hz is red.

2. Microwaves used in cooking have frequency 2.5×10^9 Hz.

3. Electromagnetic radiation of frequency 3.0×10^{13} Hz can be used for thermal imaging of buildings.

Take $c = 3.0 \times 10^8 \text{ m s}^{-1}$.

A none of them

B 1 only

C 2 only

D 3 only

E 2 and 3 only

WORKED SOLUTION

Statement 1 is false: a higher visible frequency corresponds to bluer/violet light, not red.

For statement 2,

$$f = \frac{c}{\lambda} = \frac{3.0 \times 10^8}{0.12} = 2.5 \times 10^9 \text{ Hz},$$

so statement 2 is true.

Statement 3 is also true: 3.0×10^{13} Hz lies in the infrared region, which is used in thermal imaging.

P-18

Series circuits and filament lamps

Answer D

QUESTION 18

A 6.0 V battery is connected in series with an 8.0Ω resistor, a filament lamp and an ammeter. The ammeter reading is 0.25 A.

Which pair gives the potential difference across the lamp and the resistance of the lamp at this operating point?

A 2.0 V, 8.0 Ω

B 4.0 V, 8.0 Ω

C 4.0 V, 16 Ω and power 0.50 W

D 4.0 V, 16 Ω and power 1.0 W

E 6.0 V, 24 Ω

WORKED SOLUTION

The resistor has p.d.

$$V_R = IR = 0.25 \times 8.0 = 2.0 \text{ V.}$$

So the lamp has

$$V_L = 6.0 - 2.0 = 4.0 \text{ V.}$$

Its resistance at this operating point is

$$R_L = \frac{V_L}{I} = \frac{4.0}{0.25} = 16 \Omega.$$

Its power is

$$P_L = V_L I = 4.0 \times 0.25 = 1.0 \text{ W.}$$

P-19

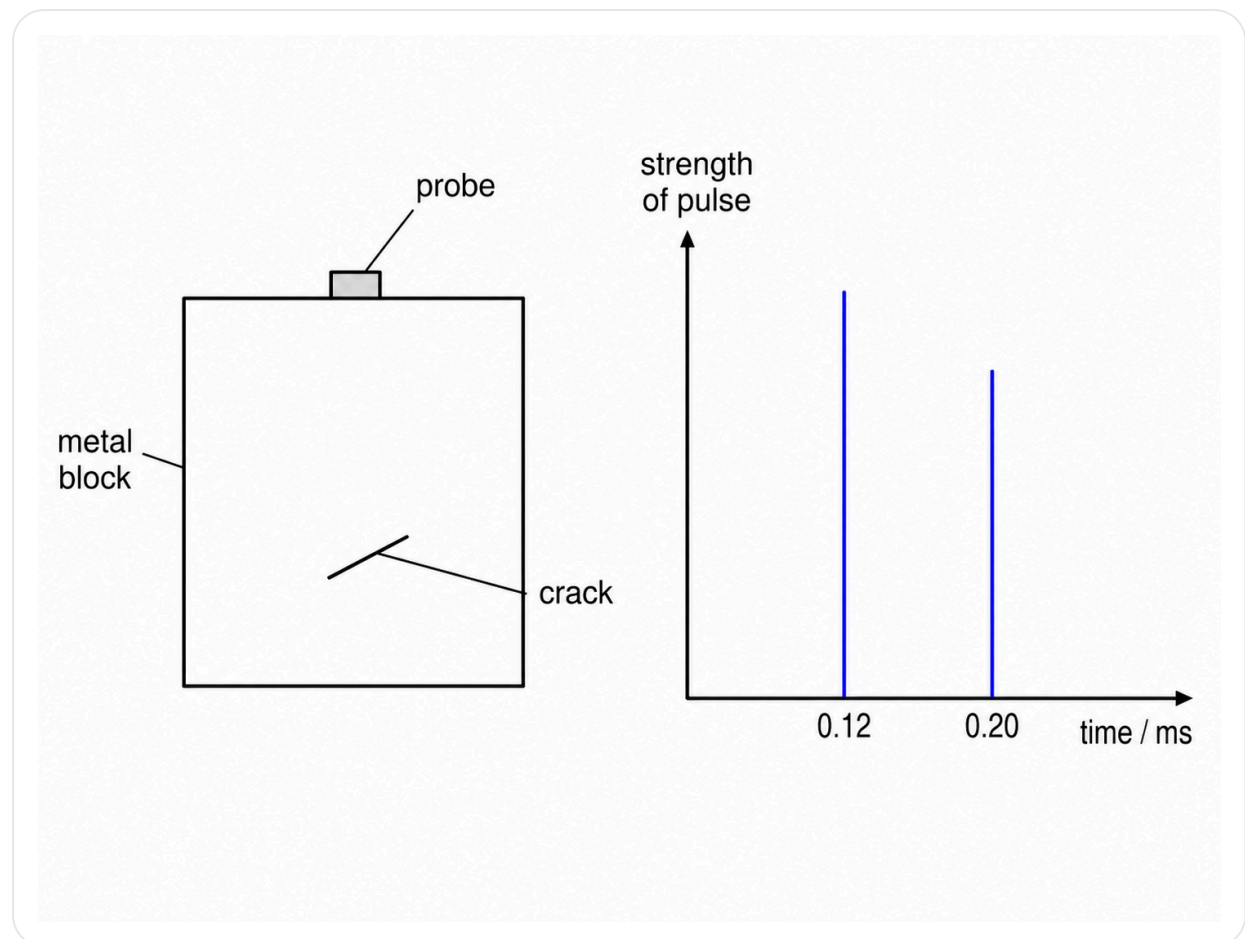
Ultrasound and echo timing

Answer C

QUESTION 19

Ultrasound is used to find a crack inside a cuboid metal block. A probe is held against the top surface and sends a short pulse into the block. Reflections return from the

crack and from the bottom surface.



The reflections return after 0.12 ms and 0.20 ms. The speed of ultrasound in the metal is 5000 m s^{-1} .

What is the ratio

$$\frac{\text{thickness of block}}{\text{depth of crack below the top surface}}?$$

A 2 : 1

B 3 : 2

C 5 : 3

D 4 : 3

E 5 : 4

WORKED SOLUTION

Echo times are round-trip times, so
depth of crack:

$$d = \frac{vt}{2} = \frac{5000 \times 0.12 \times 10^{-3}}{2} = 0.30 \text{ m.}$$

Thickness of block:

$$h = \frac{5000 \times 0.20 \times 10^{-3}}{2} = 0.50 \text{ m.}$$

Therefore

$$h : d = 0.50 : 0.30 = 5 : 3.$$

P-20

Kinematics and minimum-time travel

Answer C

QUESTION 20

A straight railway line between two cities is 60 km long. A train starts from rest, accelerates uniformly at 1.5 m s^{-2} to a maximum speed of 120 m s^{-1} , then later brakes uniformly at 2.0 m s^{-2} to come to rest at the second city.

What is the minimum possible journey time?

A 430 s

B 500 s

C 570 s

D 640 s

E 860 s

WORKED SOLUTION

Acceleration phase:

$$t_1 = \frac{120}{1.5} = 80 \text{ s}, \quad s_1 = \frac{120^2}{2(1.5)} = 4800 \text{ m.}$$

Braking phase:

$$t_2 = \frac{120}{2.0} = 60 \text{ s}, \quad s_2 = \frac{120^2}{2(2.0)} = 3600 \text{ m.}$$

Total distance used in speeding up and slowing down:

$$4800 + 3600 = 8400 \text{ m.}$$

Remaining cruising distance:

$$60000 - 8400 = 51600 \text{ m.}$$

Cruising time:

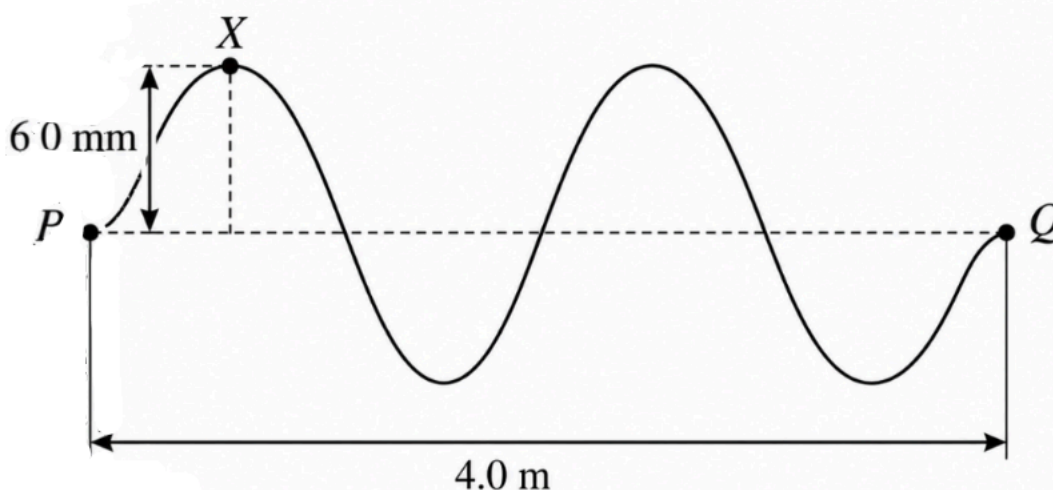
$$t_3 = \frac{51600}{120} = 430 \text{ s.}$$

Total time:

$$80 + 60 + 430 = 570 \text{ s.}$$

QUESTION 21

A transverse wave on a string has speed 500 m s^{-1} . The horizontal distance between points P and Q shown is 4.0 m .



At time $t = 0$, point X is at its maximum displacement of 6.0 mm above equilibrium.

What is the displacement of point X at $t = 7.0 \text{ ms}$, and in which direction is it moving at that instant?

- A** 6.0 mm above equilibrium, moving downward
- B** 0 mm , moving upward
- C** 0 mm , moving downward
- D** 6.0 mm below equilibrium, moving upward
- E** between 0 and 6.0 mm below equilibrium, moving downward

From the diagram, the distance from P to Q spans two complete wavelengths.
Therefore

$$2\lambda = 4.0 \text{ m},$$

so

$$\lambda = 2.0 \text{ m}.$$

The frequency of the wave is therefore

$$f = \frac{v}{\lambda} = \frac{500}{2.0} = 250 \text{ Hz}.$$

Hence the period is

$$T = \frac{1}{f} = \frac{1}{250} = 0.004 \text{ s} = 4.0 \text{ ms}.$$

Now

$$7.0 \text{ ms} = 1.75T.$$

This is equivalent to $0.75T$ after one complete cycle. Starting from maximum positive displacement, after $0.75T$ the point is at equilibrium and moving upward.

Therefore the displacement is

0 mm, moving upward.

P-22

Radioactivity and half-life

Answer B

QUESTION 22

A radioactive nuclide X decays in a single stage to a stable nuclide R. Another radioactive nuclide Y decays in a single stage to a stable nuclide S.

When a rock formed, it contained equal numbers of atoms of X, Y, R and S.

The half-life of X is T years and the half-life of Y is 2T years.

What is the value of

$\frac{\text{number of atoms of R}}{\text{number of atoms of S}}$ at a time $4T$ years after the rock formed?

A $\frac{17}{20}$

B $\frac{31}{28}$

C $\frac{5}{4}$

D $\frac{6}{5}$

E 2

WORKED SOLUTION

Let each initial number be N .

For X : after $4T$, four half-lives have passed, so

$$X = \frac{N}{16}.$$

Thus $\frac{15N}{16}$ atoms have decayed into R , giving

$$R = N + \frac{15N}{16} = \frac{31N}{16}.$$

For Y : after $4T$, two half-lives have passed, so

$$Y = \frac{N}{4}.$$

Hence $\frac{3N}{4}$ atoms have decayed into S , giving

$$S = N + \frac{3N}{4} = \frac{7N}{4}.$$

Therefore

$$\frac{R}{S} = \frac{31N/16}{7N/4} = \frac{31}{28}.$$

P-23

Waves and relative motion

Answer **D**

QUESTION 23

A ship travels into a wave that is moving in the opposite direction to the ship. The ship has speed 8.0 m s^{-1} and the wave has speed 3.0 m s^{-1} .

The front of the ship rises and falls with a time period of 8.0 s .

What are the wavelength of the wave and the period of the wave for a stationary buoy?

- A** 24 m, 8.0 s
- B** 40 m, 13.3 s
- C** 64 m, 21.3 s
- D** 88 m, 29.3 s
- E** 88 m, 8.0 s

WORKED SOLUTION

Relative speed between ship and wave crests is

$$8.0 + 3.0 = 11.0 \text{ m s}^{-1}.$$

So the wavelength is

$$\lambda = v_{\text{rel}} T_{\text{ship}} = 11.0 \times 8.0 = 88 \text{ m}.$$

For a stationary buoy, the wave period is

$$T = \frac{\lambda}{v_{\text{wave}}} = \frac{88}{3.0} = 29.3 \text{ s.}$$

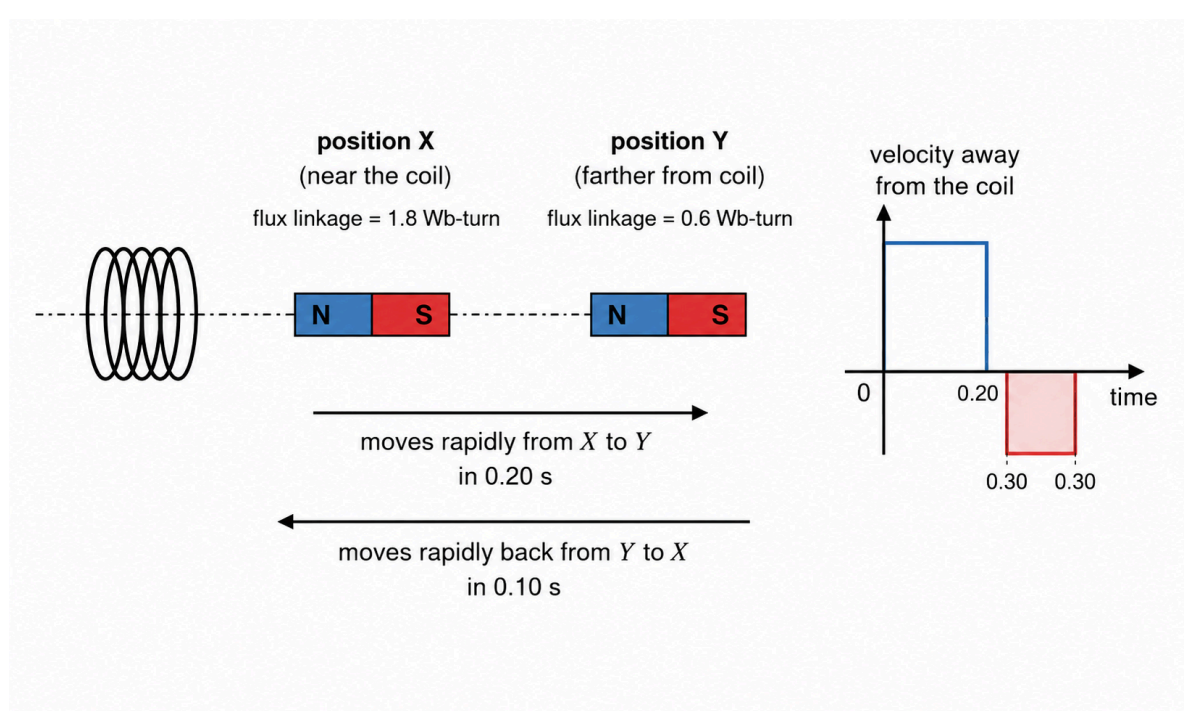
P-24

Electromagnetic induction

Answer D

QUESTION 24

A bar magnet is placed at position X close to one end of a coil and on its axis. It is moved rapidly to a more distant position Y, and then back to X.



The flux linkage of the coil is 1.8 Wb-turn when the magnet is at X and 0.6 Wb-turn when it is at Y. The motion from X to Y takes 0.20 s, while the return motion from Y to X takes 0.10 s.

What is the greatest magnitude of the induced e.m.f. during the complete motion?

A 3.0 V

B 6.0 V

C 9.0 V

D 12 V

E 18 V

WORKED SOLUTION

The change in flux linkage between X and Y is

$$\Delta(N\Phi) = 1.8 - 0.6 = 1.2 \text{ Wb-turn.}$$

For the outward motion, the average induced e.m.f. magnitude is

$$E_1 = \frac{1.2}{0.20} = 6.0 \text{ V.}$$

For the return motion, the same flux-linkage change happens in 0.10 s, so

$$E_2 = \frac{1.2}{0.10} = 12 \text{ V.}$$

The greatest magnitude is 12 V.

P-25

Seismic waves

Answer **B**

QUESTION 25

When an earthquake occurs, both P-waves and S-waves are produced at the same time and travel along the same path. P-waves travel at 5.0 km s^{-1} and S-waves travel at 3.0 km s^{-1} .

A seismic monitoring station detects the P-waves 30 s before it detects the S-waves.

How long after the earthquake does the P-wave reach the station?

A 30 s

B 45 s

C 60 s

D 75 s

E 90 s

WORKED SOLUTION

Let the distance to the station be d km. Then

$$\frac{d}{3.0} - \frac{d}{5.0} = 30.$$

So

$$d \left(\frac{1}{3} - \frac{1}{5} \right) = 30 \Rightarrow d \left(\frac{2}{15} \right) = 30 \Rightarrow d = 225 \text{ km}.$$

The P-wave travel time is therefore

$$t_P = \frac{225}{5.0} = 45 \text{ s}.$$

P-26

Pressure, ideal gas law and geometry

Answer C

QUESTION 26

A tall smooth cylinder contains air at atmospheric pressure $1.00 \times 10^5 \text{ Pa}$. The density of the air is 1.20 kg m^{-3} . A heavy piston of mass 50.0 kg and cross-sectional area 0.0200 m^2 is placed on top and allowed to fall slowly until it rests in equilibrium.

The initial height of the trapped air column is 0.50 m . Assume the air behaves as an ideal gas and its temperature remains constant. Take $g = 10 \text{ N kg}^{-1}$.

By how far does the piston fall?

A 0.040 m

B 0.080 m

C 0.10 m

D 0.125 m

E 0.20 m

WORKED SOLUTION

The piston adds pressure

$$\frac{mg}{A} = \frac{50.0 \times 10}{0.0200} = 2.5 \times 10^4 \text{ Pa.}$$

So the final pressure is

$$1.00 \times 10^5 + 2.5 \times 10^4 = 1.25 \times 10^5 \text{ Pa.}$$

At constant temperature, pV is constant, so volume decreases by a factor

$$\frac{V_f}{V_i} = \frac{p_i}{p_f} = \frac{1.00}{1.25} = 0.80.$$

The cross-sectional area is constant, so the height also decreases by a factor of 0.80:

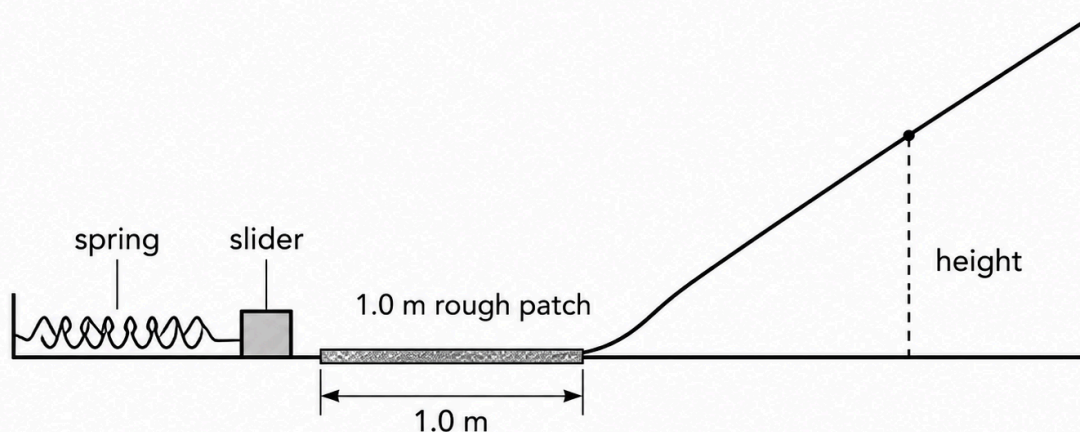
$$h_f = 0.80 \times 0.50 = 0.40 \text{ m.}$$

Therefore the piston falls by

$$0.50 - 0.40 = 0.10 \text{ m.}$$

QUESTION 27

A small slider of mass 30 g is at rest near the bottom of a frictionless slope and in contact with a light uncompressed spring as shown.



In one test, compressing the spring by 5.0 cm sends the 30 g slider up the slope to a maximum vertical height of 20 cm.

The slider is then replaced by a 20 g slider. The spring is compressed by 15 cm. During the motion, the slider must first cross a rough horizontal patch of length 1.0 m on which a constant resistive force of 0.20 N acts. Elsewhere resistance is negligible.

To what maximum vertical height does the new slider rise? Take $g = 10 \text{ N kg}^{-1}$.

A 120 cm

B 150 cm

C 170 cm

D 180 cm

E 270 cm

WORKED SOLUTION

From the first test, spring energy equals gain in gravitational potential energy:

$$\frac{1}{2}k(0.050)^2 = 0.030 \times 10 \times 0.20.$$

So

$$0.00125k = 0.060 \Rightarrow k = 48 \text{ N m}^{-1}.$$

With the new slider and compression 0.15 m, spring energy is

$$E_s = \frac{1}{2} \times 48 \times (0.15)^2 = 0.54 \text{ J}.$$

Work done against the rough patch is

$$W_f = 0.20 \times 1.0 = 0.20 \text{ J}.$$

So energy left for gravitational potential energy is

$$0.54 - 0.20 = 0.34 \text{ J}.$$

For the 20 g = 0.020 kg slider,

$$mgh = 0.34 \Rightarrow h = \frac{0.34}{0.020 \times 10} = 1.7 \text{ m} = 170 \text{ cm}.$$

Biology

27 QUESTIONS • COMPLETE SOLUTIONS

Preserved Biology Harder than ESAT source set. Question wording, option order, answer and solution method are unchanged.

B-01

Cells • surface area to volume

Answer C**QUESTION 1**

Two unicellular organisms, X and Y, are approximately cube-shaped.

The side length of Y is twice the side length of X.

Both organisms have:

- the same oxygen demand per unit volume;
- the same maximum rate of oxygen diffusion per unit surface area.

Compared with X, by what factor is the **oxygen demand per unit surface area** of Y greater?

A 1/2**B** 1**C** 2**D** 4**E** 8**WORKED SOLUTION**

For a cube, volume is s^3 and surface area is $6s^2$.

$$\text{oxygen demand per unit surface area} \propto s^3 / 6s^2 = s/6$$

Doubling the side length therefore doubles the oxygen demand per unit surface area.

Answer: 2.

B-02

Movement across membranes

Answer **F**

QUESTION 2

A section of dialysis tubing contains a solution of glucose and starch. It is placed in a large beaker of pure water.

The membrane is permeable to water and glucose, but not to starch.

Consider the statements while the system is approaching equilibrium.

1. While the glucose concentration remains higher inside the tubing than outside, there is a net movement of glucose out of the tubing.
2. Starch can leave the tubing by osmosis because its concentration is higher inside.
3. Even after the glucose concentrations have become equal, water may still show a net movement into the tubing because starch remains inside.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Glucose can cross the membrane, so while its concentration is higher inside, its net movement is outward by diffusion.

Osmosis describes the net movement of **water**, not starch, so statement 2 is false.

Because starch cannot cross the membrane, it can continue to lower the water potential inside the tubing. Water may therefore continue to enter even after glucose has equilibrated.

Statements 1 and 3 only are correct.

B-03

Diffusion and active transport

Answer B

QUESTION 3

A root-hair cell contains nitrate ions at a concentration of **4 mmol dm⁻³**.

A respiratory poison is added which completely stops active transport, but the cell membrane remains permeable to nitrate ions.

The cell is placed separately into three soil solutions:

soil solution	nitrate concentration / mmol dm ⁻³
X	1
Y	4
Z	10

Which row gives the expected **net passive movement of nitrate ions immediately after active transport stops?**

A X: into cell; Y: no net movement; Z: out of cell

B X: out of cell; Y: no net movement; Z: into cell

C X: out of cell; Y: into cell; Z: into cell

D X: no net movement; Y: no net movement; Z: into cell

E X: out of cell; Y: out of cell; Z: into cell

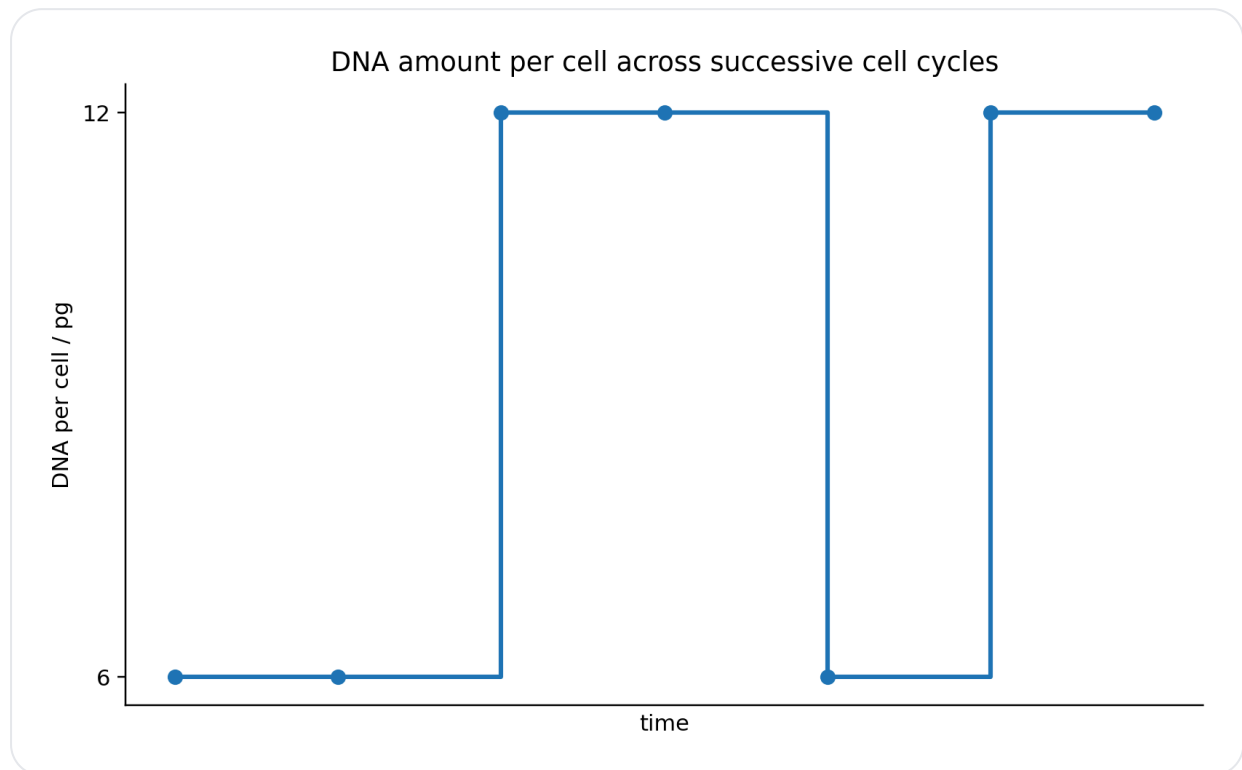
WORKED SOLUTION

With active transport disabled, only passive movement down the concentration gradient remains.

- X: 1 outside versus 4 inside → net movement **out**.
- Y: 4 outside versus 4 inside → **no net movement**.
- Z: 10 outside versus 4 inside → net movement **in**.

Answer B.

QUESTION 4



A diploid species has **12 chromosomes** in each body cell.

Immediately after mitosis, one cell contains **6 pg** of DNA.

It passes through another cell cycle, as represented by the first rise and fall on the graph. Both daughter cells then replicate their DNA and are observed at the **second high plateau**, immediately before their next mitosis.

Which row is correct at this second high plateau?

A 1 cell; 12 total chromosomes; 12 pg total DNA

B 2 cells; 24 total chromosomes; 12 pg total DNA

C 2 cells; 24 total chromosomes; 24 pg total DNA

D 2 cells; 48 total chromosomes; 24 pg total DNA

E 4 cells; 48 total chromosomes; 24 pg total DNA

WORKED SOLUTION

After the first mitosis there are two cells. Each has 12 chromosomes and 6 pg DNA.

DNA replication doubles DNA quantity but does not change chromosome number.

So immediately before the next mitosis, each of the two cells has 12 chromosomes and 12 pg DNA.

across both cells: 24 chromosomes and 24 pg DNA

Answer C.

B-05

Meiosis and sex determination

Answer C

QUESTION 5

A male mammal has a diploid chromosome number of $2n = 10$ and an XY sex-chromosome system.

One diploid germ cell completes meiosis normally and produces four sperm.

Across **all four sperm together**, which row is correct?

A 8 total autosomes; 4 total sex chromosomes; X:Y sperm = 2:2

B 16 total autosomes; 2 total sex chromosomes; X:Y sperm = 1:1

C 16 total autosomes; 4 total sex chromosomes; X:Y sperm = 2:2

D 20 total autosomes; 4 total sex chromosomes; X:Y sperm = 1:3

E 16 total autosomes; 8 total sex chromosomes; X:Y sperm = 2:2

WORKED SOLUTION

Each sperm is haploid and therefore contains 5 chromosomes: 4 autosomes plus 1 sex chromosome.

$$4 \text{ sperm} \times 4 \text{ autosomes} = 16 \text{ autosomes}; 4 \text{ sperm} \times 1 \text{ sex chromosome} = 4 \text{ sex chromosomes}$$

Normal meiosis of an XY germ cell gives equal numbers of X-bearing and Y-bearing sperm, so the ratio is 2:2.

Answer C.

B-06

Inheritance · conditional probability

Answer D

QUESTION 6

In an animal species:

- allele A is dominant and produces black fur in genotype Aa;
- genotype aa produces white fur;
- genotype AA is lethal before birth.

Two **living black animals** mate.

What is the probability that **exactly one of their next two live offspring is white**?

A 1/9

B 2/9

C 1/3

D 4/9

E 2/3

WORKED SOLUTION

A living black animal cannot be AA because AA is lethal, so both parents are Aa. Before lethality the cross gives 1/4 AA, 1/2 Aa and 1/4 aa. Among **live** offspring, the probabilities therefore become:

$$P(\text{black} \mid \text{alive}) = 2/3 \quad P(\text{white} \mid \text{alive}) = 1/3$$

Exactly one white among two live offspring can occur in either order:

$$2 \times (1/3) \times (2/3) = 4/9$$

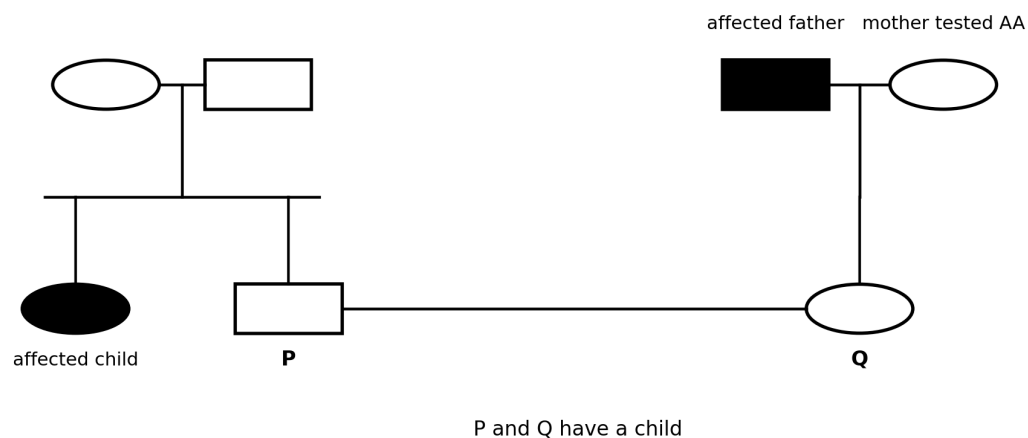
Answer D.

B-07

Pedigrees · conditional probability

Answer C

QUESTION 7



Filled symbol = affected by the same autosomal recessive condition

The same autosomal recessive condition occurs in both families shown.

In the family on the left, two unaffected parents have an affected child and an unaffected son, P.

In the family on the right, Q's father is affected and Q's unaffected mother has been genetically tested and is homozygous dominant.

P and Q have a child.

What is the probability that their child is affected?

A 1/12

B 1/8

C 1/6

D 1/4

E 1/3

WORKED SOLUTION

P's parents must both be Aa because they are unaffected but have an affected aa child.

Given that P is unaffected, P has a $2/3$ probability of being Aa.

Q's father is aa and Q's mother is AA, so Q must be Aa.

If P is a carrier, $Aa \times Aa$ gives an affected child with probability $1/4$.

$$(2/3) \times (1/4) = 1/6$$

Answer C.

B-08

DNA base pairing

Answer D

QUESTION 8

One strand of a DNA molecule has the following base composition:

base	percentage
A	18
T	30
C	22
G	30

The complete double-stranded DNA molecule contains **2000 bases**.

How many **guanine bases** are present in the complete molecule?

A 440

B 480

C 500

D 520

E 600

WORKED SOLUTION

On the complementary strand, the percentage of G equals the percentage of C on the first strand, so it is 22%.

Across the two equal-length strands, the overall fraction of G is therefore:

$$(30\% + 22\%) / 2 = 26\%$$

$$0.26 \times 2000 = 520$$

Answer D.

QUESTION 9

The following coding-DNA triplets have the stated meanings.

triplet	amino acid
GCT	alanine
GCC	alanine
GTT	valine
GAT	aspartic acid
CCT	proline

A gene originally contains **GCT-GAT-CCT**.

Two mutations occur separately:

X: GCC-GAT-CCT

Y: GTT-GAT-CCT

1. Mutation X must alter the amino-acid sequence.
2. Mutation Y alters one amino acid in the encoded sequence.
3. Mutation Y must necessarily produce a visible change in phenotype.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

GCT and GCC both code for alanine, so mutation X does not alter this amino acid.

GTT codes for valine rather than alanine, so mutation Y changes one amino acid.

An amino-acid change does not guarantee a visible phenotypic change.

Statement 2 only is correct — answer C.

B-10

Gene technologies

Answer B

QUESTION 10

A bacterial strain is initially sensitive to kanamycin.

Scientists use a plasmid containing a functional kanamycin-resistance gene. The plasmid is cut at a site **outside** this resistance gene.

A gene of interest is cut with the same restriction enzyme, mixed with the plasmid and DNA ligase, and the bacteria are transformed.

The bacteria are then grown on agar containing kanamycin. Some colonies grow.

Assume that no new mutation conferring kanamycin resistance occurs during the experiment.

Which conclusion is justified?

A Every growing colony must contain the gene of interest.

B Every growing colony must have obtained DNA carrying a functional kanamycin-resistance gene, but growth alone does not show whether the gene of interest was inserted.

C Every bacterium exposed to the plasmid must have been transformed.

D The gene of interest must have entered the bacterial chromosome rather than a plasmid.

E DNA ligase must have cut the plasmid DNA.

WORKED SOLUTION

The original bacteria are kanamycin-sensitive. A colony that grows on kanamycin must therefore have acquired functioning resistance under the conditions described.

However, a cut plasmid can potentially re-form without receiving the desired insert. Kanamycin survival therefore does not prove that the gene of interest is present.

Answer B.

B-11

Stem cells

Answer B

QUESTION 11

Three human cell populations have these properties.

population	observed ability
P	can give rise to all cell types required to form a complete organism
Q	can differentiate into any body-cell type but cannot by itself form a complete organism
R	obtained from adult bone marrow and can form several types of blood cell

A patient requires replacement blood cells.

Which row is correct?

- A** P pluripotent; Q totipotent; R multipotent; least broad sufficient population = P
- B** P totipotent; Q pluripotent; R multipotent; least broad sufficient population = R
- C** P totipotent; Q multipotent; R pluripotent; least broad sufficient population = Q
- D** P pluripotent; Q multipotent; R totipotent; least broad sufficient population = R
- E** P multipotent; Q pluripotent; R totipotent; least broad sufficient population = P

WORKED SOLUTION

P is totipotent because it can generate all cell types needed for a complete organism.

Q is pluripotent because it can generate all body-cell types but not a complete organism.

R is multipotent because it can form a restricted range of related cell types.

For replacement blood cells, R is the least developmentally broad population that could be sufficient.

Answer B.

B-12

Variation · genes and environment

Answer E

QUESTION 12

Two genetically different clonal lines, A and B, are grown under two nitrate conditions. All other measured conditions are kept the same.

clonal line	low nitrate: mean height / cm	high nitrate: mean height / cm
A	12	20
B	16	24

1. Environmental conditions can affect height.
2. Genetic differences could contribute to differences in height.
3. These data prove that nitrate availability is the only environmental factor that can affect plant height.

Which conclusions are supported?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Within each clonal line, height changes when nitrate availability changes, supporting an environmental effect.

Clonal line B is taller than clonal line A under both nitrate conditions, which is consistent with a genetic contribution.

The experiment does not establish that nitrate is the only environmental factor capable of affecting height.

Statements 1 and 2 only are supported — answer E.

B-13

Enzymes · experimental interpretation

Answer **F**

QUESTION 13

An enzyme is tested at three temperatures and three pH values. The table shows the time required to produce the **same fixed quantity of product**.

	20°C	40°C	60°C
pH 4	180 s	60 s	>300 s
pH 6	120 s	30 s	240 s
pH 8	200 s	75 s	>300 s

1. Of the conditions tested, pH 6 and 40°C gives the highest reaction rate.
2. These results prove that the enzyme's exact optimum is pH 6 and exactly 40°C.
3. At pH 6, the relative reaction rate at 40°C is four times that at 20°C.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

For a fixed quantity of product, rate is inversely proportional to the time taken.

The shortest measured time is 30 s at pH 6 and 40°C, so statement 1 is true.

$$\text{relative rate ratio} = (1/30) / (1/120) = 4$$

Statement 3 is therefore true.

The exact optimum could lie between the values tested, so statement 2 is not justified.

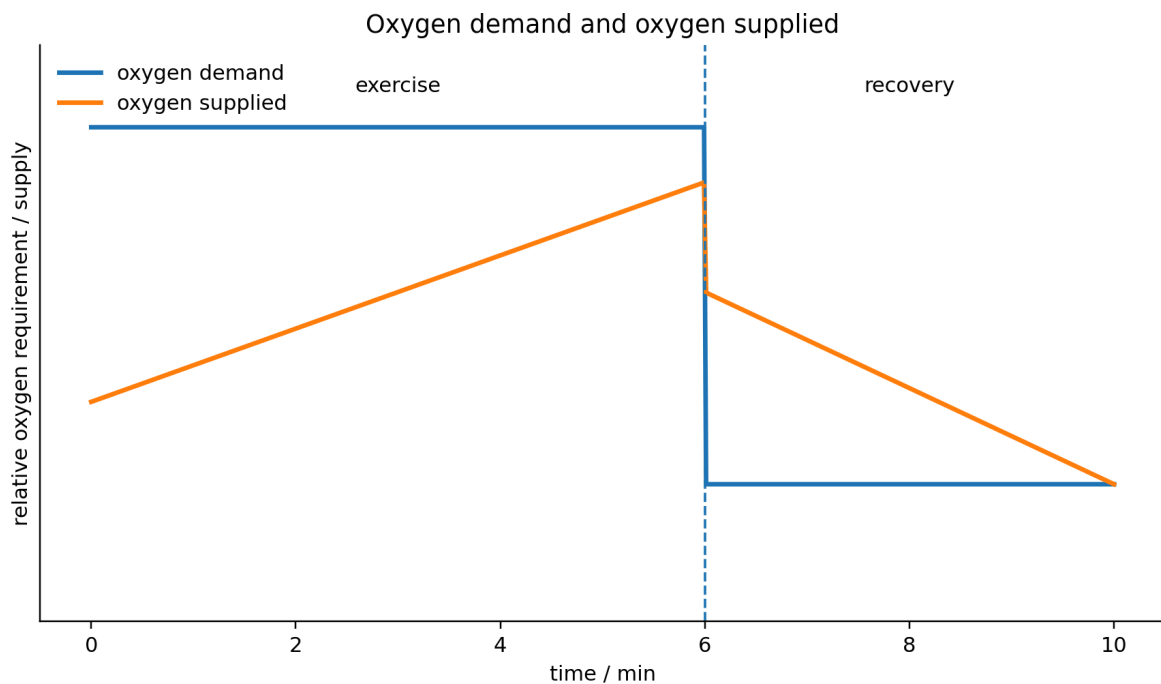
Answer F: 1 and 3 only.

B-14

Respiration during exercise

Answer E

QUESTION 14



The graph shows relative oxygen demand and oxygen supplied during six minutes of exercise followed by recovery.

1. During the exercise period, the graph is consistent with anaerobic respiration contributing to energy release.
2. During the early recovery period, oxygen supply exceeding the current oxygen demand is consistent with repayment of oxygen debt.
3. When oxygen supply and demand become equal at the end of recovery, aerobic respiration must have stopped.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

During exercise, oxygen demand exceeds the oxygen supplied, so aerobic respiration alone cannot account for the full stated demand. This is consistent with anaerobic respiration contributing.

During early recovery, oxygen supply remains above the current demand, which is consistent with repayment of oxygen debt.

Equal supply and demand at the end does not mean respiration has stopped; it means oxygen supply is meeting the current requirement.

Statements 1 and 2 only — answer E.

B-15

Gas exchange · quantitative diffusion

Answer C

QUESTION 15

A simple cubic organism has side length **s mm**.

Its cells consume oxygen at a rate equivalent to **1 unit per mm³ per minute**.

Oxygen can cross the entire external surface at a maximum rate of **0.5 units per mm² per minute**.

The organism has no internal transport system.

What is the largest value of s for which diffusion across the surface could supply enough oxygen?

A 1 mm

B 2 mm

C 3 mm

D 4 mm

E 6 mm

WORKED SOLUTION

Oxygen demand equals the volume:

$$\text{demand} = s^3$$

Surface area is $6s^2$, so maximum supply is:

$$0.5 \times 6s^2 = 3s^2$$

Enough oxygen is supplied when $3s^2 \geq s^3$. For $s > 0$ this gives $s \leq 3$.

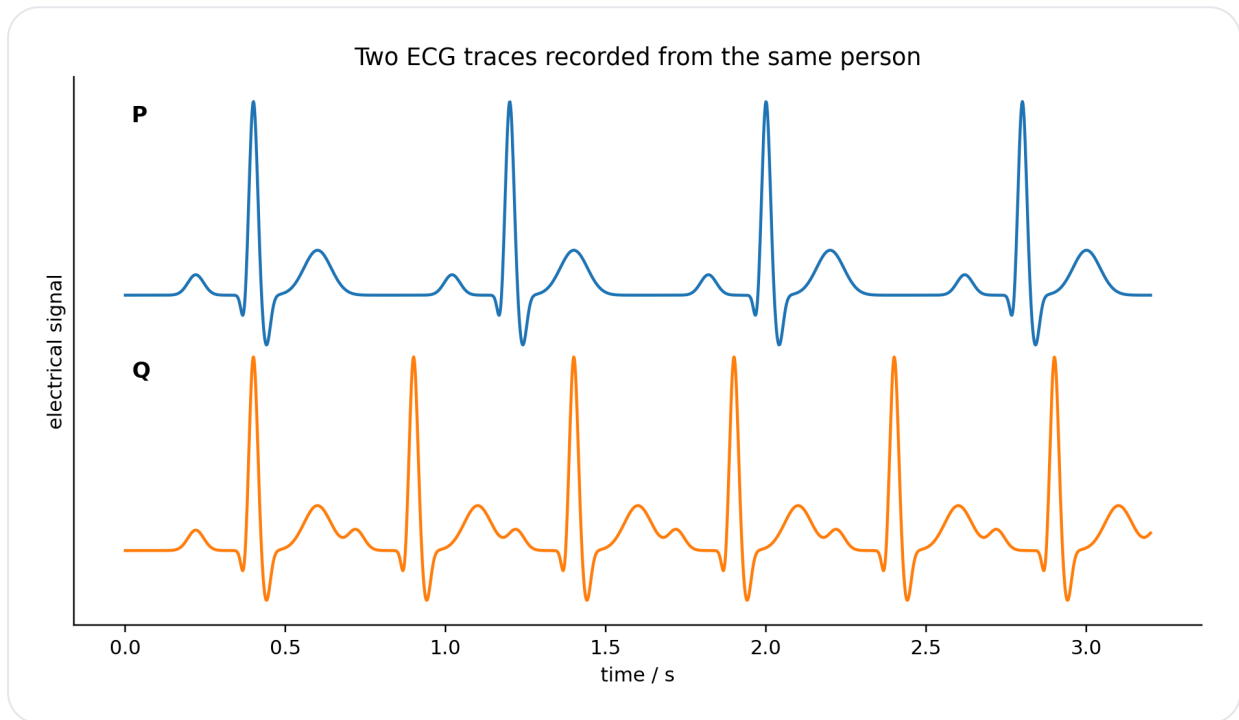
Largest possible side length = 3 mm — answer C.

B-16

ECG · heart rate and cardiac output

Answer **D**

QUESTION 16



Two ECG traces are recorded from the same person.

The spacing between corresponding large peaks is approximately **0.80 s in P** and **0.50 s in Q**.

Assume the volume of blood pumped per heartbeat is unchanged.

Which row is correct?

A P = 60 min⁻¹; Q = 75 min⁻¹; blood/minute ratio Q:P = 1.25

B P = 75 min⁻¹; Q = 100 min⁻¹; ratio = 1.33

C P = 75 min⁻¹; Q = 120 min⁻¹; ratio = 1.50

D P = 75 min⁻¹; Q = 120 min⁻¹; ratio = 1.60

E $P = 80 \text{ min}^{-1}$; $Q = 120 \text{ min}^{-1}$; ratio = 1.50

WORKED SOLUTION

$$P \text{ heart rate} = 60 / 0.80 = 75 \text{ beats min}^{-1}$$

$$Q \text{ heart rate} = 60 / 0.50 = 120 \text{ beats min}^{-1}$$

If stroke volume is unchanged, blood pumped per minute changes in the same ratio:

$$120 / 75 = 1.60$$

Answer D.

B-17

Kidney · blood composition

Answer B

QUESTION 17

The concentrations of four substances are measured in two vessels connected to the same organ.

substance	vessel P	vessel Q
oxygen	9	6
carbon dioxide	5	7
urea	7	2
glucose	5	5

Values are in arbitrary but comparable units.

Which interpretation best fits the data?

A The organ is a lung; P is a pulmonary vein and Q a pulmonary artery.

B The organ is a kidney; P could be a renal artery and Q a renal vein.

- C** The organ is a liver; P must be a hepatic vein and Q a hepatic artery.
- D** The organ is a kidney; P must be the renal vein and Q the renal artery.
- E** The organ is skeletal muscle; P must be a vein and Q an artery.

WORKED SOLUTION

From P to Q, oxygen falls and carbon dioxide rises, consistent with respiration by the organ's cells.

Urea falls sharply, indicating removal from the blood. This strongly fits a kidney.

Blood enters a kidney in a renal artery and leaves through a renal vein, so P could be the renal artery and Q the renal vein.

Answer B.

B-18

Digestion · bile and lipase

Answer B

QUESTION 18

Equal amounts of lipid and lipase are used in four experiments.

tube	lipid	bile	pH	time to fixed endpoint
A	large droplets	absent	8	120 s
B	mechanically emulsified into small droplets	absent	8	45 s
C	large droplets	present	8	50 s
D	large droplets	present	2	>300 s

Which conclusion is best supported?

- A** Bile is itself the enzyme that hydrolyses lipid.

B In these experiments, an important effect of bile could be increasing lipid surface area, because mechanical emulsification produces a similar increase in rate.

C Lipase has its optimum at pH 2.

D Bile converts lipid directly into glucose.

E Lipase functions only when bile is present.

WORKED SOLUTION

Tubes B and C reach the endpoint in very similar times even though B has no bile. In B the lipid has instead been mechanically broken into small droplets.

This supports the idea that an important effect of bile is emulsification, which increases the lipid surface area available to lipase.

Tube D also shows that an unsuitable low pH greatly reduces the reaction rate.

Answer B.

B-19

Homeostasis · ADH

Answer E

QUESTION 19

A drug prevents ADH from increasing the water permeability of the collecting ducts but does not directly affect filtration or glucose reabsorption.

A dehydrated person takes the drug.

Compared with a healthy dehydrated person, which changes are expected?

1. A greater urine volume.
2. A lower concentration of urea in the urine is likely because more water remains in the urine.
3. Glucose must appear in the urine because ADH is absent from the collecting duct.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

ADH normally increases water permeability of the collecting ducts, increasing water reabsorption.

If that effect is blocked, less water is reabsorbed, so urine volume is greater and dissolved urea is likely to be less concentrated.

ADH is not the mechanism responsible for glucose reabsorption, so glucose does not have to appear in the urine.

Statements 1 and 2 only — answer E.

B-20

Nervous and hormonal coordination

Answer **E**

QUESTION 20

A person's hand touches a very hot surface.

The hand is rapidly withdrawn. A few seconds later, adrenaline contributes to an increased heart rate.

1. Withdrawal of the hand can be coordinated through a reflex pathway involving a relay neurone in the central nervous system.
2. Neurotransmitter molecules can diffuse across synapses in the reflex pathway.
3. Adrenaline travels from an endocrine gland to the heart along motor neurones.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

A rapid withdrawal reflex can involve a relay neurone in the central nervous system.

At chemical synapses, neurotransmitter molecules diffuse across the synaptic gap.

Adrenaline is a hormone and is transported in the blood, not along motor neurones.

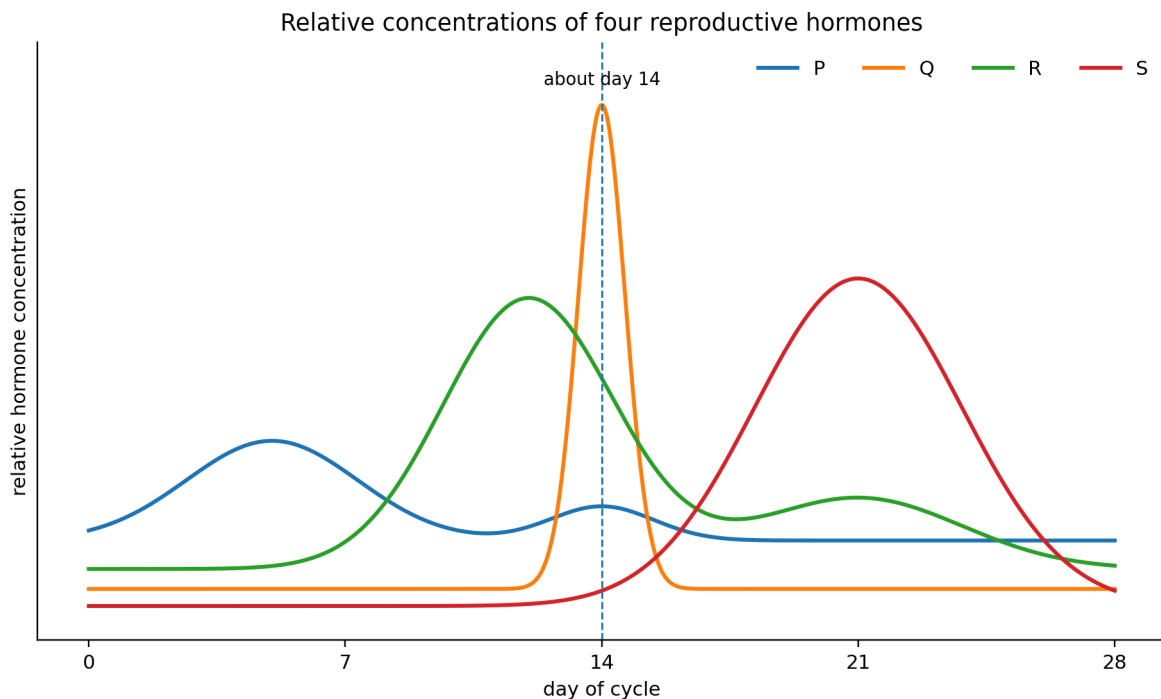
Statements 1 and 2 only — answer E.

B-21

Reproductive hormones

Answer A

QUESTION 21



The graph shows four hormones, P, Q, R and S.

The large narrow peak around day 14 represents the hormone that triggers ovulation.
The broad peak during the second half of the cycle represents the hormone that maintains the uterine lining.

Which pair of curves represents the **two pituitary hormones whose production tends to be inhibited when oestrogen and progesterone are maintained at high levels by combined hormonal contraception?**

A P and Q

B P and R

C Q and S

D R and S

E Q and R

WORKED SOLUTION

The sharp mid-cycle peak is LH, so Q is LH.

The broad second-half peak is progesterone, so S is progesterone.

The remaining pituitary hormone is FSH, represented by P.

The two pituitary reproductive hormones are therefore P and Q.

Answer A.

B-22

Clinical trials · evidence

Answer E

QUESTION 22

A medicine and a placebo are tested on two similar groups.

group	people tested	people whose symptoms improved	people reporting side effects
medicine	200	130	20
placebo	200	90	8

1. The improvement rate is 20 percentage points higher in the medicine group.
2. The proportion reporting side effects is 2.5 times as high in the medicine group as in the placebo group.
3. The trial proves that every improvement in the medicine group was caused by the medicine.

Which are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

medicine improvement = $130/200 = 65\%$

placebo improvement = $90/200 = 45\%$

The difference is 20 percentage points.

side-effect ratio = $(20/200) / (8/200) = 10\% / 4\% = 2.5$

However, improvement also occurred in the placebo group, so the trial does not show that every individual improvement in the medicine group was caused by the medicine.

Statements 1 and 2 only — answer E.

B-23

Ecology · quadrat sampling

Answer **D**

QUESTION 23

Ten randomly positioned **0.50 m × 0.50 m** quadrats are used to sample a plant population.

quadrat	count
1	0
2	4
3	7
4	0
5	5
6	2
7	8
8	0
9	3
10	1

The habitat has an area of **100 m²**.

Which row gives both the estimated population and the **frequency of occurrence** of the plant?

A 300; 70%

B 600; 30%

C 1200; 30%

D 1200; 70%

E 4800; 70%

WORKED SOLUTION

Total organisms counted = 30, so the mean is 3 per quadrat.

Each quadrat has area 0.25 m², giving a density of:

$$3 / 0.25 = 12 \text{ plants m}^{-2}$$

$$\text{estimated population} = 12 \times 100 = 1200$$

The species occurs in 7 of the 10 quadrats, so its frequency is 70%.

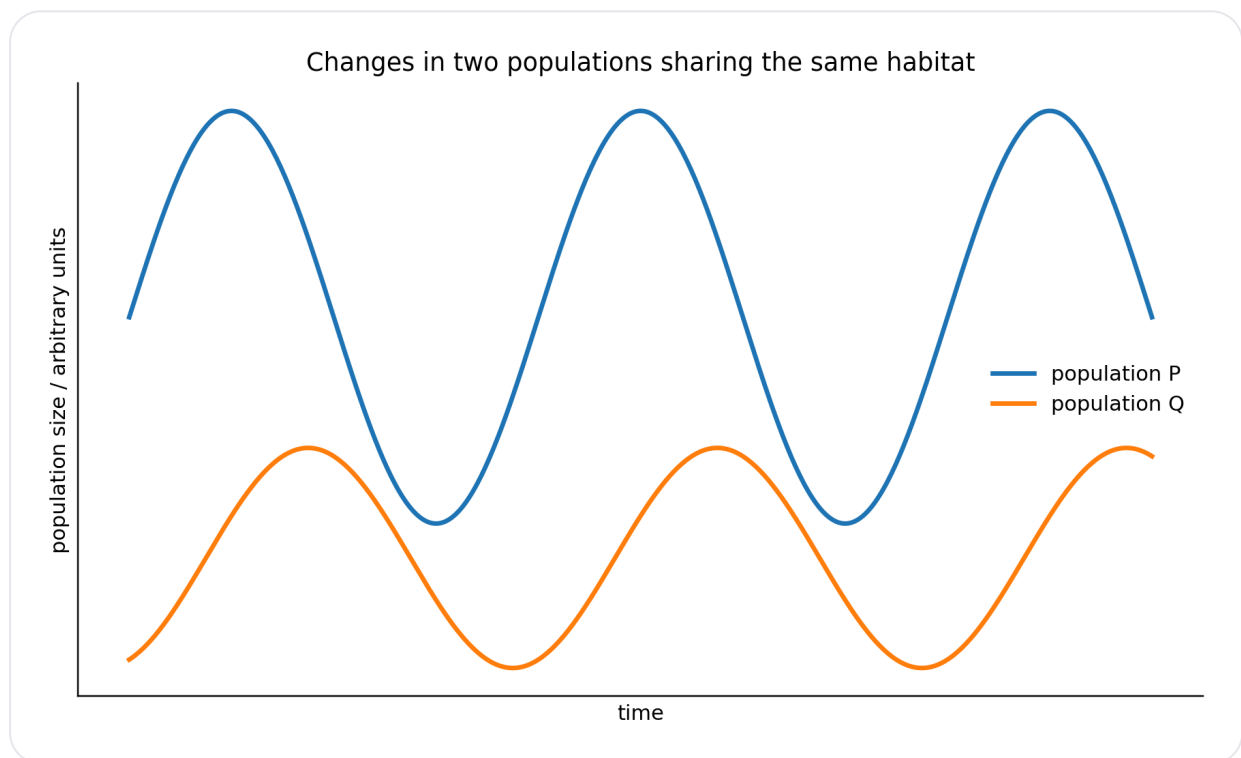
Answer D.

B-24

Ecology · population interactions

Answer F

QUESTION 24



Two populations in the same habitat show the changes in the graph.

Population Q tends to peak after population P.

1. The pattern is consistent with Q being a predator whose population partly depends on P as prey.
2. The graph alone proves that changes in P are the sole cause of changes in Q.
3. If P is an important prey species for Q, a decline in P could later contribute to a decline in Q.

Which statements are justified?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

The delayed peaks are consistent with a possible predator-prey relationship.

If P is important prey, reduced P availability could later reduce Q.

However, the graph alone cannot prove that P is the sole cause of changes in Q because other biotic or abiotic factors may also affect Q.

Statements 1 and 3 only — answer F.

B-25

Carbon cycle · decomposition

Answer E

QUESTION 25

Equal masses of leaf litter containing radioactive carbon, ^{14}C , are placed in three sealed experimental systems.

The amount of $^{14}\text{CO}_2$ released after one week is measured.

system	conditions	$^{14}\text{CO}_2$ produced / arbitrary units
A	non-sterile soil, 25°C	60
B	sterilised soil, 25°C	5
C	non-sterile soil, 5°C	20

1. Living decomposers contribute substantially to carbon release from the litter.
2. Lower temperature can reduce the rate at which carbon is returned from the litter as carbon dioxide.
3. These data show that light is essential for decomposition.

Which conclusions are supported?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

A produces far more labelled CO_2 than sterilised B, supporting a major role for living decomposers.

A and C both contain decomposers, but the lower-temperature system releases much less labelled CO_2 , supporting a temperature effect.

Light was not manipulated, so statement 3 is not supported.

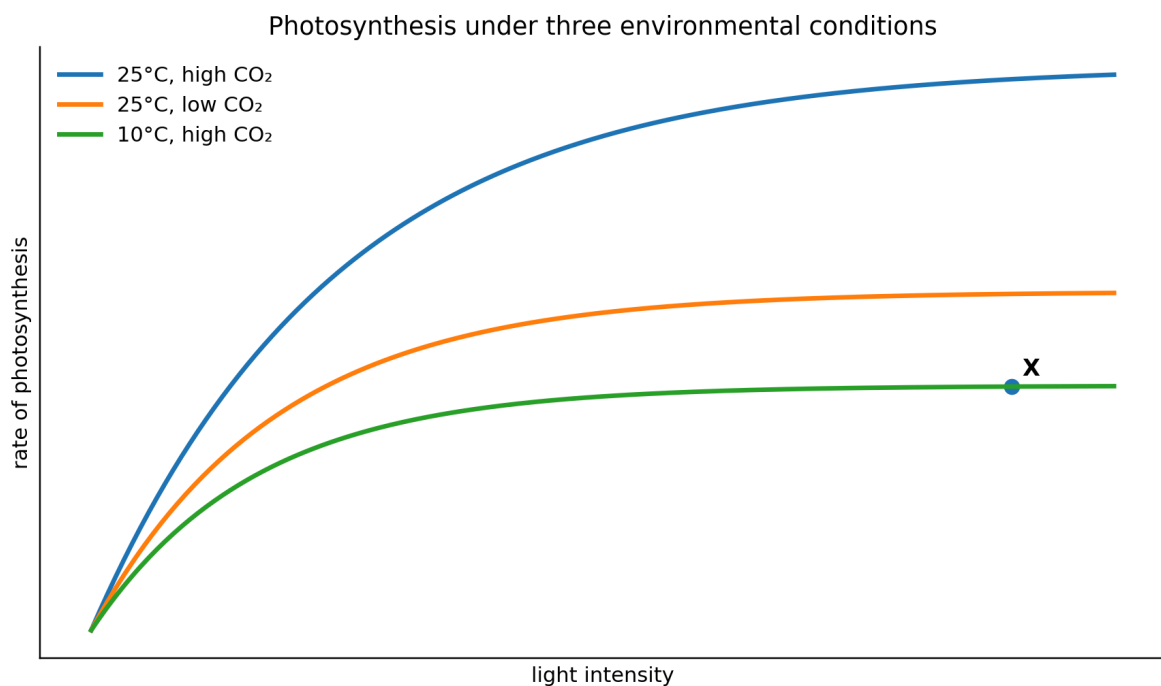
Statements 1 and 2 only — answer E.

B-26

Photosynthesis · limiting factors

Answer C

QUESTION 26



The graph shows photosynthesis under three conditions.

Point X is at high light intensity, high carbon dioxide concentration and **10°C**.

Which single change is most strongly supported by the graph as likely to increase the rate at X?

- A** Increase the light intensity further.
- B** Increase the carbon dioxide concentration further.
- C** Increase the temperature from 10°C towards 25°C.
- D** Decrease the carbon dioxide concentration.

E Decrease the temperature.

WORKED SOLUTION

At X, the 10°C high-CO₂ curve is already near its plateau, so increasing light further is unlikely to have much effect.

Carbon dioxide is already high.

At the same high light and high carbon dioxide concentration, the 25°C curve reaches a much higher rate. The graph therefore supports temperature as the key limiting factor at X.

Answer C.

B-27

Plant physiology · photosynthesis and translocation

Answer H

QUESTION 27

A mature leaf is supplied with radioactive carbon dioxide, ¹⁴CO₂, for a short period.

In one plant, a ring of tissue is removed from the stem below the leaf. The treatment removes the **phloem but leaves the xylem intact**.

A third plant has an intact stem but its treated leaf is kept in darkness.

treatment	radioactivity in sugars in/above treated leaf	radioactivity in roots	relative water uptake
illuminated, intact stem	1000	400	6
illuminated, phloem removed	1300	30	6
dark leaf, intact stem	50	10	5

1. The much greater incorporation of ¹⁴C in illuminated leaves is consistent with carbon dioxide being incorporated into sugars during photosynthesis.
2. The large reduction in radioactivity reaching the roots when phloem is removed supports the role of phloem in sugar transport.

3. The maintenance of water uptake after phloem removal is consistent with xylem remaining functional.

Which conclusions are supported?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

The illuminated intact leaf incorporates far more labelled carbon into sugars than the dark leaf, consistent with carbon dioxide fixation during photosynthesis.

Removing phloem reduces root radioactivity from 400 to 30 while labelled sugar accumulates in or above the treated region, supporting phloem translocation.

Water uptake remains at 6 after phloem removal, consistent with the intact xylem continuing to transport water.

All three conclusions are supported — answer H.

source sets. This solution book is ready to support the later construction of the matching full test.